Recurring transit-like signals in six TESS fields: evidence, source ambiguity, and limits of validation

Richard Higgins
richard@relh.net

20 October 2026
Preprint draft; scientific evidence assessed through 13 October 2026

Abstract

We report an exploratory search and assessment of six transit-like signals in public TESS photometry. The strongest recurring signals are associated with the catalog fields of TIC 80387959 and TIC 402358350, at investigated periods of 4.19217 and 11.76954 days. A clock fixed before inspecting two later epochs of TIC 80387959 recovers dimming of 621 ± 98 and 651 ± 114 parts per million (ppm). Retrospective local-baseline fits retain positive depths in six sectors. For TIC 402358350, eight usable events remain positive under twelve local-baseline choices, with combined depths of 590–659 ppm. Its later-sector timing uncertainty fails the declared prediction gate. Neither signal has secure source attribution: neighboring stars can supply the missing flux, image models fail shallow known-host controls, and conditional astrophysical scenario calculations do not meet the joint validation thresholds. The other four signals retain single-epoch physical degeneracy, weak or timing-dependent recurrence, or evidence favoring contamination. We provide the adopted clocks, event and processing comparisons, control failures, alternative explanations, and a linked evidence inventory with file hashes. The overlapping searches and selected injections do not establish survey completeness or calibrated false-alarm rates. These results identify follow-up questions rather than validated planets, and catalog non-matches do not establish novelty.

1 Introduction

Public photometry from the Transiting Exoplanet Survey Satellite (TESS; Ricker et al. 2015) allows searches for repeated dimming beyond the mission's original target lists. A recurring dip is an observable pattern; its astrophysical interpretation requires identifying the star that dims and distinguishing a planetary transit from stellar eclipses, blended variability, and instrumental effects. Repeating a measurement with another reduction of the same exposures tests processing sensitivity but does not supply independent observing evidence.

This paper documents six unresolved signals retained from several related searches. We prioritize TIC 80387959 and TIC 402358350 because their dimming persists across multiple epochs and analysis choices. We also retain the unfavorable evidence that prevents a planetary conclusion. The remaining fields—TIC 241928959, TIC 469309982, TIC 96166003, and TIC 5546257—illustrate distinct limitations rather than four additional validated discoveries. The designation "ambiguous planetary candidate" (APC), used in the accompanying research archive, is an internal working category. It is neither an official TESS Object of Interest designation nor a probability statement. TIC 5546257 is retained for its instructive contamination evidence, while its on-target planet interpretation is deprioritized.

Our contribution is an auditable record of the investigated signals and decision rules. It includes failed predictions, unusable observations, negative controls, and superseded interpretations. The six signals were not all selected from one homogeneous survey, so their number cannot be divided by a single target count to estimate occurrence. No signal is claimed as a newly validated or confirmed planet. Methods and control outcomes appear in Section 3, the two priority fields in Section 4, and the remaining fields in Section 7.

2 Data and signal inventory

We use public TESS-SPOC and Quick-Look Pipeline (QLP) products obtained through MAST (Caldwell et al., 2020; Huang et al., 2020). The saved run records identify the product, sector, flux column, quality selection, timestamp convention, retrieval location, and raw-content SHA-256. Simple-aperture photometry (SAP) and presearch-data-conditioning photometry (PDCSAP, abbreviated PDC below) are reductions of the same exposures. Likewise, QLP and SPOC products for an overlapping sector share observing photons. They are useful processing comparisons, not independent astronomical confirmations. QLP extraction changed from sector 94 (Petitpas et al., 2026); an apparent inter-sector depth change cannot be assigned a physical origin without addressing extraction, sampling, and dilution.

Table 1 summarizes the examined epochs and a representative later or local measurement. An "opened" sector has already been inspected in this investigation; the label does not establish adequate coverage, successful recovery, or an independent holdout. The sector 64 and 99 cutouts requested for TIC 80387959 contain collateral pixels at the target aperture. They are unusable target observations, not transit non-detections. Metadata-only footprints for TIC 96166003/71 and TIC 5546257/8/63 remain untested and are excluded from the evidence counts.

Table 1: Retained signals and representative evidence. Errors are conditional on the stated photometric model. A positive fixed-clock depth alone does not satisfy timing or source-attribution requirements.

TIC	Examined usable sectors	Representative evidence	Principal limitation
80387959	10,36,37,63,90,100	Frozen later-sector depths 621 ± 98, 651 ± 114 ppm	Host and depth differences unresolved
402358350	40,41,54,81	Eight local-baseline events; 590–659 ppm combined	Later timing gate fails; host uncertain
241928959	92	Five events; local depth 11992 ± 786 ppm	One epoch; grazing/companion degeneracy
469309982	9,10,36,37,63,64,90	Frozen sector 64: 76 ± 26 ppm; sector 90 primary: -54 ± 32 ppm	Mixed, clock-dependent recurrence
96166003	19,43,44,59,86	Frozen sector 59 primary: 265 ± 52 ppm	Alternative clocks disagree; timing fails
5546257	9,35,36,62,89,99	Frozen sector 62: 386 ± 30 ppm; neighbor Gaia dimming	Contamination favored; exact alias unresolved

Catalog identities and positions are given in Table 2. Gaia DR3 coordinates are ICRS at J2016.0 (Gaia Collaboration et al., 2023); observations at other dates require propagation where appropriate. TIC stellar parameters (Stassun et al., 2019) characterize the catalog stars, without establishing that those stars host the dips. We deliberately do not report adopted planet radii or masses. Normal Gaia renormalized unit weight error (RUWE) and absent non-single-star solutions do not exclude unresolved companions. The elevated RUWE of TIC 241928959 is a concern, not a companion detection.

Table 2: Catalog context for the named target stars. Coordinates are Gaia DR3 at J2016.0; T magnitudes, temperatures and radii are TIC values. Symmetric errors summarize the stored catalog fields and do not encompass all stellar-model systematics.

TIC	Gaia DR3 source ID	RA (deg)	Dec (deg)	T	$T_{\rm eff}$ (K)	$R_\star/R_\odot$	RUWE
80387959	5362389543781401216	165.310575	−48.613881	10.524	5220 ± 125	0.797 ± 0.045	0.828
402358350	4318715445843296256	295.150929	+15.640412	9.432	6079 ± 138	0.995 ± 0.046	1.000
241928959	6865254022910168320	297.986934	−21.503148	12.614	3396 ± 157	0.651 ± 0.020	3.356
469309982	5259582389096040960	148.847729	−56.685744	9.215	5445 ± 125	0.879 ± 0.047	0.978
96166003	161856980718462208	72.588533	+33.707024	10.423	5171 ± 237	0.825 ± 0.049	0.983
5546257	5426593115802740992	143.087875	−40.474512	9.588	5826 ± 131	0.925 ± 0.045	0.913

Table 3: Adopted reference clocks, rounded for display. BTJD is $\mathrm{BJD_{TDB}} - 2457000$. Values identify historical predictions, not unique orbital solutions or parameter precision. Full stored values and alternative hypotheses remain in the linked evidence. Durations are fixed box widths, not physical transit-duration posteriors.

TIC	P (days)	t_0 (BTJD)	D (days)	Training sectors
80387959	4.192168718	1575.375831	0.092736	10,36,37,63
402358350	11.769544332	2392.992869	0.144000	40,41,54
241928959	3.532592898	3805.411657	0.032000	92
469309982	2.890464274	1545.740130	0.144000	9,10,36,37
96166003	0.955530474	1816.967196	0.059520	19,43,44
5546257	1.767290230	1545.965336	0.133056	9,35,36

3 Search, evidence preservation, and controls

3.1 Observations and search populations

We searched public TESS photometry for recurring transit-like dimming Ricker et al. (2015), using TESS-SPOC and Quick-Look Pipeline (QLP) products Caldwell et al. (2020); Huang et al. (2020). Target properties and initial contamination estimates came from the TESS Input Catalog (TIC; Stassun et al. (2019)). The six retained signals emerged from several successive, overlapping campaigns. They are not six discoveries from a single uniform survey, and the denominators in Table 4 must not be added to obtain a unique-star count. The investigation was adaptive: later experiments responded to earlier controls and candidate assessments. We report the executed selection and its limitations rather than interpreting the final shortlist as a statistical planet yield.

Table 4: Completed search scopes. Counts describe processing outcomes, not astrophysical classifications. The weak-seed stars were already counted in earlier searches; other scopes also overlap.

Scope	Attempts	Reconciled outcome
Solar-radius pool	7,004	4,481 below threshold; 2,494 flagged; 28 screening passes, all reviewed; one original data error.
Weak-seed refinement	689	688 measurements and one error; 23 local flags, all reviewed; 22 deprioritized.
Faint QLP pilot	100	46 below threshold; 54 flagged; no errors or full screening passes.
Faint QLP expansion	1,845	950 below threshold; 888 flagged; seven original data failures. Six recoveries gave 1,844 usable searches; 17 peak follow-ups on 16 stars.

The solar-radius inventory selected catalog radii of 0.81–1.10 $R_\odot$, TESS magnitudes 6–12, and

distances up to 120 pc. Of 16,581 opportunities with early and later TESS-SPOC coverage, 408 were excluded by exact identifiers in saved planet, candidate, and eclipsing-binary catalogs, 1,635 failed the contamination-ratio requirement of at most 0.2 or lacked its value, and 7,534 lacked an adjacent early-sector pair. The remaining 7,004 were ranked by $R_\star^2 10^{0.2(T-10)}/\sqrt{N_{\rm train}}$. This is a metadata-based noise proxy, not measured quietness. The earliest contiguous group of two or three sectors numbered at most 55 trained discovery, while the latest available ordinary sector numbered 56–107 was reserved. No temperature cut was applied to this pool.

The fainter QLP pool used $12 < T \leq 13.5$, catalog radii 0.15–0.5 $R_\odot$, distances up to 120 pc, $T_{\rm eff} \leq 4000$ K, and the same contamination-ratio limit. Dated catalog exclusions, TIC quality dispositions, and prior raw-data access were checked before selection. From 38,822 eligible metadata targets, 1,945 had a suitable four-sector sequence and a separate reserved epoch. Discovery used the latest four consecutive QLP sectors ending no later than sector 96; reserved sectors were at least eight sector numbers from every training sector. Their actual training baseline could not exceed 120 days. Sector-number separation alone does not establish independent or usable observations. The four-sector design followed limited known-planet controls and was therefore an informed design change, not a blind survey choice. The first 100 targets formed the pilot and the remaining 1,845 the expansion. Six expansion failures were recovered with three usable assigned sectors; one target remained unsearchable with only two. Recoveries are neither additional stars nor equivalent to four-sector coverage.

3.2 Preprocessing and recurring-dip search

For TESS-SPOC we used corrected PDCSAP flux and its supplied errors. We retained quality-zero samples with finite times and positive, finite fluxes and uncertainties, sorted timestamps, and removed duplicate timestamps. FITS identities, integrity information, raw-product URLs, retrieval records, and SHA-256 hashes were retained. Times are expressed as BTJD = $\mathrm{BJD_{TDB}} - 2457000$. QLP flux and error columns were recorded per product. Where a target's explicitly enabled fallback encountered no usable published errors, it assigned equal errors from the robust adjacent-difference scale, $1.4826\ \mathrm{MAD}(\Delta f)/\sqrt{2}$. This estimates short-timescale scatter rather than a calibrated correlated-noise uncertainty. QLP discovery normally used inverse-variance 10-minute bins; native sampling was retained for the solar-radius search and QLP refinements.

Each sector was divided at gaps longer than 0.3 days. We divided flux and errors by a running-median trend with a 1.5-day window and repeated the analysis with a 2.5-day window. Segments shorter than the filter window or 30 samples were excluded, and half a window was trimmed from each edge. Errors were floored at the robust first-difference scatter of the normalized flux. Only extreme upward excursions, above eight such scatter units, were removed; downward outliers were preserved. At least 200 clean samples were required. These operations can attenuate or remove transits, especially near gaps; a different reduction of the same exposures supplies a robustness test, not independent photons.

The multi-sector search used Astropy box least squares (BLS; Kovács et al. (2002); Astropy Collaboration et al. (2022)) with a likelihood objective, periods from 0.5 days to $\min(20\,\mathrm{days}, B/2.5)$ for baseline B, and durations $D = 0.04$, 0.06, 0.09, 0.13, and 0.18 days. The BLS frequency factor was 2. The ordinary dense-search baseline limit was 100 days, extended to 120 days for the four-sector QLP experiment. Up to three peaks separated by more than 2% in period were retained, including harmonics for later inspection. Formal depth signal-to-noise ratio (SNR) of at least 7 triggered diagnostics; a full screening pass required SNR at least 8 and all additional gates below. Historical single-sector searches instead used periods 0.7–$\min(10, B/2.5)$ days, durations 0.06, 0.09, 0.13, 0.18, and 0.24 days, and frequency factor 0.8.

For a clock (P, t_0, D), in-event samples satisfied $|\phi| < D/2$ and local comparison samples $D < |\phi| < \min(3D, P/2)$, where ϕ is the wrapped time offset. Depth was the inverse-variance-weighted comparison mean minus the event mean; its conditional variance was the sum of the two mean variances. Measurements required at least two in-event and five comparison samples. An event was supported when its formal depth SNR was at least 2. Odd/even depths, a half-phase secondary, inverted-flux BLS, sinusoidal variability, gap proximity, event consistency, and alternate detrending were checked. These are heuristic screening diagnostics, not calibrated false-alarm probabilities.

The weak-seed experiment revisited 689 below-threshold stars with at least two unopened additional early sectors, retaining at most nine early sectors and the original strongest peak as its sole seed. A local review flag required SNR at least 8 under both detrendings, periods agreeing within 0.5%, and neither timing profile touching its search boundary. This was a restricted refinement, distinct from full-domain rediscovery or a full screening pass.

3.3 Frozen clocks and the holdout record

Before testing reserved photometry, we recorded training inputs, the refined clock, all accepted timing-profile pairs, and decision rules. Local refinement tested durations $0.8D$, D, and $1.2D$, retaining period/epoch pairs within $\Delta \log \mathcal{L} = 4.5$ of the maximum, including disconnected islands. The period half-width was $3DP/B_{\mathrm{seed}}$; grids contained at least 2,049 and at most 50,001 points. The epoch floor was $\max(D/50, D/\max[\mathrm{SNR}, 1])$. These are conditional box-model profiles with a heuristic floor, not physical or correlated-noise confidence intervals.

We aligned profile epochs to a common cycle and propagated them to the reserved observations. The maximum timing displacement plus three epoch floors had to be less than $D/2$, the accepted profile could not reach the period-grid boundary, and training required three supported events. Recovery additionally required combined fixed-clock SNR at least 5 and two supported reserved events. With adequate timing and expected formal SNR at least 5, observed SNR below 2 constituted a non-recovery; insufficient coverage, sensitivity, or timing otherwise remained inconclusive. Fixed-period shifted-phase comparisons were dependent controls, not survey false-alarm estimates. Refitted clocks, enlarged windows, alternate reductions, and retrospective season omissions were retained as exploratory analyses; already inspected observations did not become fresh holdouts.

One operational exception is material. TIC 5546257 sector 62 was opened after its expanded-training prediction was frozen but before the parent solar-radius queue's terminal audit. This violated that queue's reserved-cache absence assertion. The queue stopped after ranks 3920–4431, the chronology and raw-data hash were preserved in a supplemental audit, and the remaining 2,572 attempts resumed under a separate frozen plan. Original target records and predictions were not overwritten. We therefore do not claim perfect compliance with every original queue holdout rule. Later audits distinguish historical pre-access cache checks from the subsequent opened-data state.

3.4 Sensitivity and conditional reliability controls

Table 5 includes unsuccessful controls as well as recoveries. Five known shallow planets were selected before their photometry was opened, using the solar-radius opportunity pool and catalog depth proxies near 200 ppm. Only Kepler-93 b and TOI-5159 b passed all gates. TOI-1435 b was below threshold, TOI-5788 b failed SNR and detrending tests, and variability obscured HD 63433 d. Recovery of the stronger TOI-5788 c does not change the preselected-planet result of two out of five. Figure 1 retains all five outcomes.

The shallow-injection experiment applied exposure-integrated 100 and 300 ppm boxes to raw flux before detrending in three preselected below-threshold targets. It tested periods 1.37, 4.73,

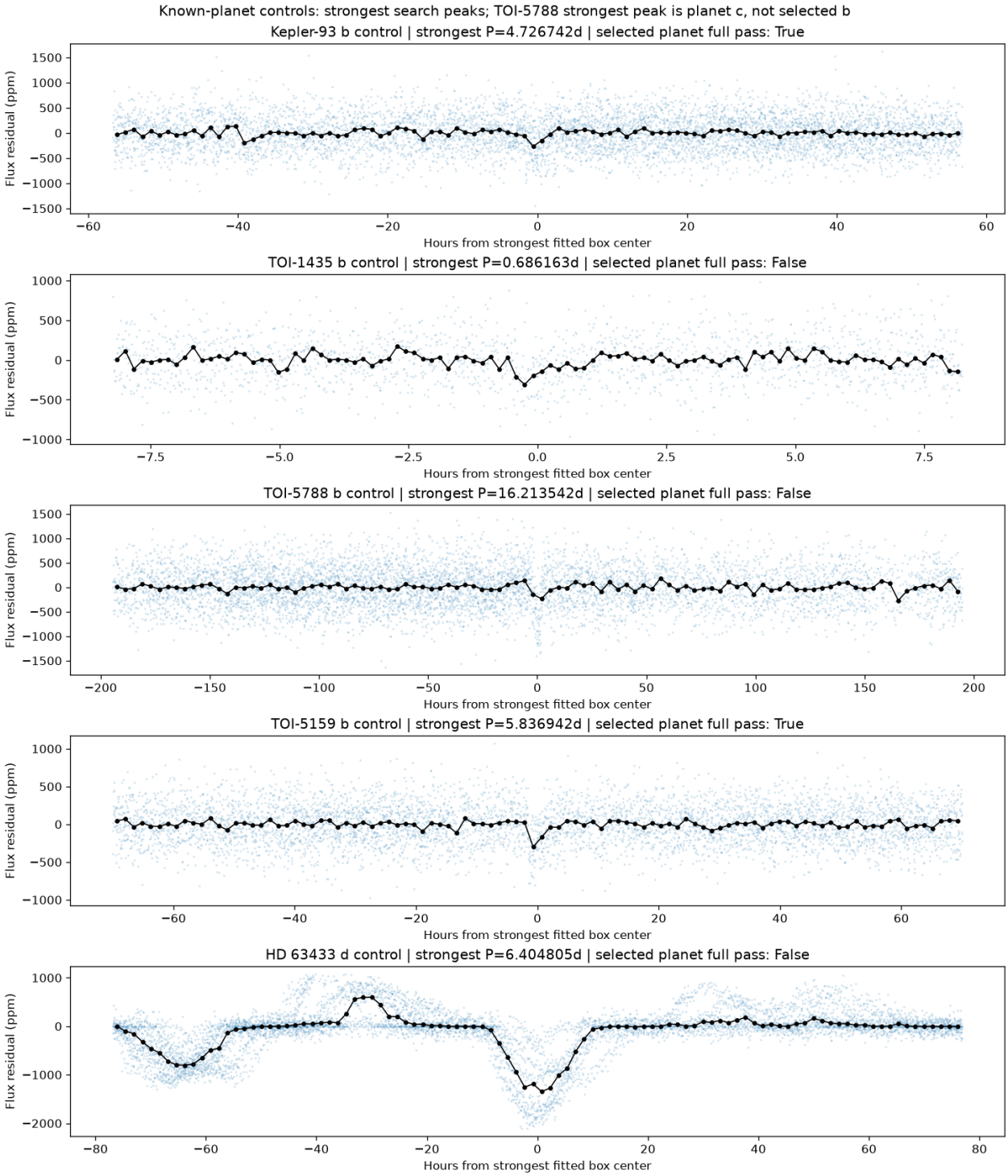


Figure 1: Unchanged-pipeline searches of five preselected known planets. The figure is retained from the archived control experiment. Successful period agreement and full screening passage are different outcomes; the two full passes do not establish population completeness.

Table 5: Selected controls and their interpretation. These small, selected experiments do not measure survey completeness or planet probabilities.

Experiment	Outcome	Principal limitation
Known shallow planets	2/5 preselected planets pass; 4/5 periods occur among the three retained peaks.	Period recovery alone is not gate passage; nonrandom sample.
Raw-flux shallow injections	3/18 recoveries at 100 ppm; 11/18 at 300 ppm, across three selected stars.	One phase per configuration, box shapes, limited periods and durations.
TIC 469309982 permutations	0/40 local-flag exceedances in each of three block models.	Original target and seed selection are conditioned on.
TIC 96166003 permutations	0/40 selected-family exceedances in each of three block models.	Uses all already-opened sectors and a selected period family.
Deprioritized signals	TIC 362222987: 46/48 shuffled maxima exceed the observation; TIC 396160833: 17/48.	Conditional maximum-power comparisons, not physical classifications.

and 13.13 days, durations 0.04 and 0.12 days, and one phase per configuration. All three 100-ppm recoveries occurred at the shortest period and longer duration; none occurred at the two longer periods. Existing matching baseline signals were not credited as recoveries. Thus negative searches cannot exclude shallow planets on the basis of these examples.

For the two lower-priority signals in Table 5, deterministic 0.25-, 0.5-, and 1-day blocks of paired raw flux and errors were permuted within sector and gap-separated segments. Forty draws per model yielded zero exceedances; each model's binomial 95% Monte Carlo interval is 0–8.81%. Counts were not pooled to claim a smaller bound. Block seams, destroyed long correlations, retained signal fragments, and unproven exchangeability limit interpretation. Four deterministic injections for TIC 469309982 passed its local flag, but one selected a competing epoch in one detrending; all four TIC 96166003 injections recovered the imposed period and epoch. These checks verify processing behavior under selected conditions, not the original search false-alarm probability.

3.5 Attribution and uncertainty limitations

Difference images compared mean local baseline and in-event pixels, with propagated diagonal errors and inverse-variance event stacking. Conditional source fits used mission response templates plus a planar background. Neither fitted registration uncertainty nor response-model mismatch was fully propagated into source rankings. Kepler-93 b exposes the practical consequence: the target was preferred in sector 40, but a neighbor in sector 41, despite a known shallow planetary signal. Stable rankings across registration or event omissions can therefore accompany wrong attribution.

A separately frozen pointing-correction control passed three of four cases; the contaminant's discrimination decreased in one sector. A template-free paired-aperture diagnostic passed five of eight cases and preferred the wrong Kepler-93 region twice. Neither met its gate for candidate application. These failures prevent converting favorable candidate image fits into calibrated host probabilities or using poor-model rankings alone to reject a planet. Conditional circular, dark-occultor fits with exposure-integrated `batman` models Kreidberg (2015) likewise assess compatibility under assumed hosts and dilution; they do not exclude stellar companions.

Time-averaging audits of the two strongest signals follow the correlated-noise rationale of Pont et al. (2006). They retain adverse event omissions and insufficient-bin cases, but use different estimators: one compares fitted residual bins with simulated white-noise refits, while the other

examines out-of-event bins. Their scalar error inflation is a sensitivity analysis, not a complete covariance model or interchangeable reliability score. Agreement between reductions sharing photons, successful arithmetic verification, and conditional astrophysical scenario weights do not provide independent observations or validation Giacalone et al. (2021).

3.6 Detailed full-screen rejection gates

For reproducibility, a full-screen peak was flagged for any of the following: formal SNR below 8; fewer than three supported events; an unmeasurable odd/even comparison or discrepancy above 3σ; an unmeasurable half-phase secondary or secondary SNR above 3; inverted-to-original maximum BLS power ratio at least 0.8; sinusoidal likelihood improvement greater than the box improvement; or duty cycle $D/P > 0.15$. Additional flags applied when duration-binned correlated noise was unmeasurable or reduced SNR below 7, event-depth consistency had conditional chi-square survival below 0.001, one event contributed more than 75% of summed positive event SNR squared, or more than half the supported events lay within $\max(0.15\,\mathrm{days}, D)$ of a clean-segment edge. The alternate detrending's strongest period had to agree within 2%, and every supplied aperture crowding fraction had to be at least 0.8. Missing crowding information remained a flag, including for QLP products; separately labeled diagnostic-only follow-ups did not erase original failures. The discovery noise ratio used robust scatter of out-of-event duration-bin means relative to their expected white errors, with a floor of one, at least 15 bins, and at least half the expected sampling in each bin. These exact numerical gates define the experiment; they do not constitute an empirically calibrated planet classifier.

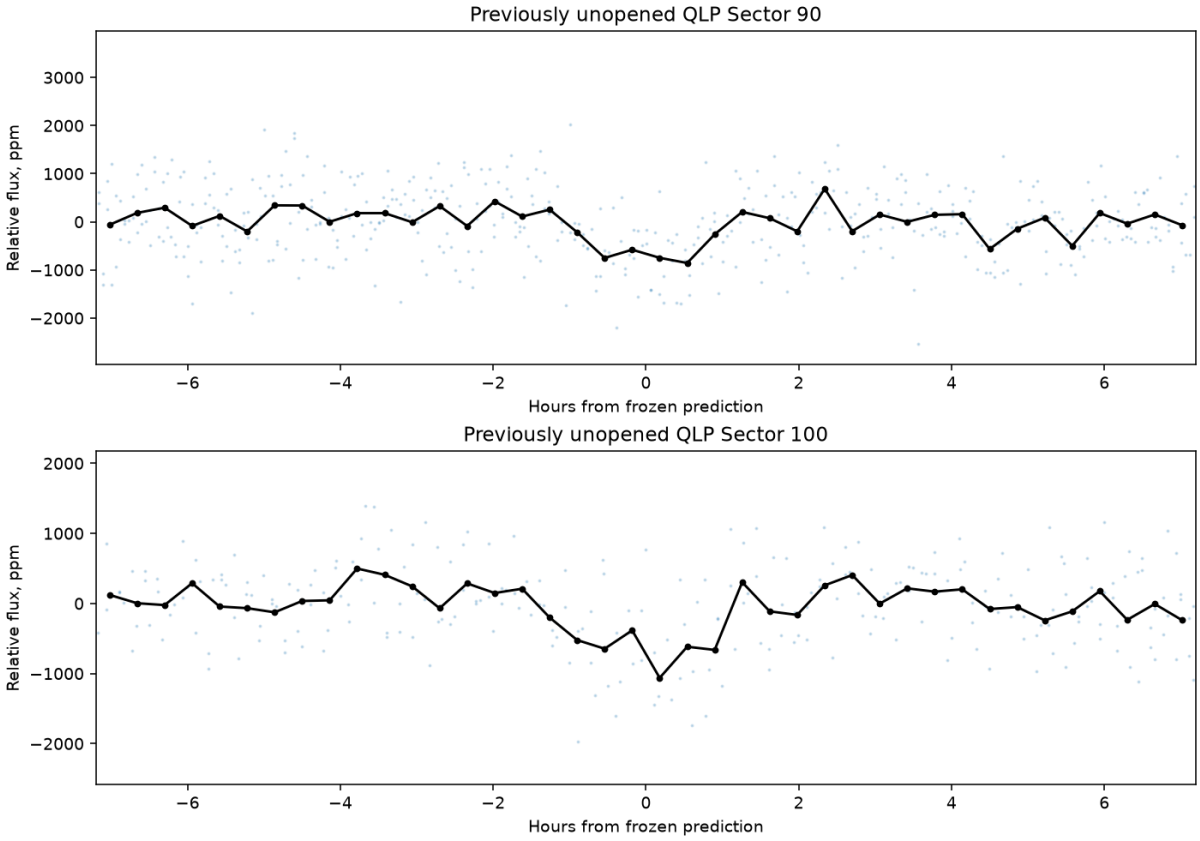


Figure 2: TIC 80387959 in QLP sectors 90 and 100 at the prediction fixed before those products were retrieved. These are the original follow-up reductions, with their original preprocessing and sampling. Positive recurrence at the fixed clock is evidence about the signal; it does not identify the star producing it. Displayed measurement errors are conditional on the adopted processing and noise model.

4 The two priority signals

4.1 TIC 80387959: successful prediction tests with an unresolved host

TIC 80387959 provides the clearest successful prediction test in this investigation. Its adopted clock has period $P = 4.192168718052227$ d, reference midpoint $t_0 = 1575.375831123655$ BTJD, and fixed box duration $D = 0.092736$ d (2.225664 h). Throughout this paper, BTJD $= \mathrm{BJD}_{\mathrm{TDB}} - 2457000$. The digits specify the saved prediction; they do not imply that the physical orbit is known to the displayed precision. Predictions based on the earlier sectors were frozen before retrieval of QLP sectors 90 and 100. Without adjusting the clock, the original follow-up procedure measured depths of 621 ± 98 and 651 ± 114 ppm, respectively (Figure 2). Five and three events entered those measurements, of which four and two passed the procedure's individual support criterion. The propagated timing spreads, 0.02477 and 0.02679 d, were below the half-duration gate of 0.046368 d. These results support recurrence at the earlier prediction, while leaving the variable source and physical explanation undetermined.

A subsequent retrospective test used native corrected flux from all six already inspected sectors, without an additional global flattening filter. It fitted an exposure-integrated box simultaneously with separate local intercepts and slopes for each event. Table 6 reports these measurements, which have different samples and nuisance models from the original prediction test. All six sectors

Table 6: TIC 80387959: retrospective joint local-baseline box fits. The final column applies a heuristic time-averaging noise inflation to the same depth estimates. All errors are conditional, not complete uncertainty budgets or calibrated search significances.

Sector	Product	Events	Depth (ppm)	Adjusted error (ppm)
10	TESS-SPOC PDCSAP	4	1038 ± 109	113
36	TESS-SPOC PDCSAP	6	813 ± 87	92
37	TESS-SPOC PDCSAP	5	904 ± 91	105
63	TESS-SPOC PDCSAP	5	807 ± 93	130
90	QLP DET_FLUX	6	613 ± 89	92
100	QLP DET_FLUX	4	582 ± 99	133

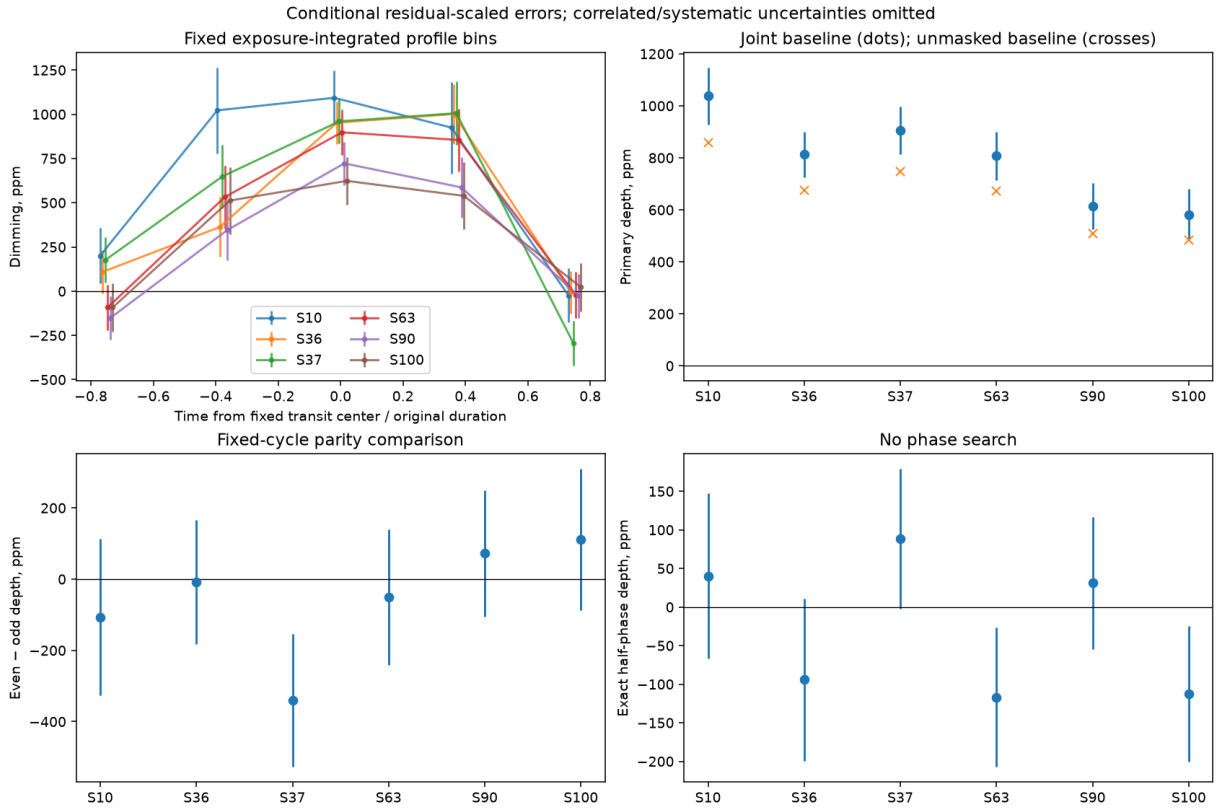


Figure 3: Six-sector fixed-clock shape, depth, parity, and half-phase tests for TIC 80387959. Native corrected-flux models jointly fit local event baselines, without additional global flattening. All data were already inspected before this retrospective analysis. Conditional errors omit a complete correlated-noise and dilution model. A positive dip across these tests does not exclude a blended stellar eclipse.

and every single-event omission retain a positive depth; the smallest omitted-event depth is 494 ppm in sector 100. Thirty event windows enter this particular analysis. Neither these windows nor alternative reductions of their exposures are thirty independent confirmations of a planetary interpretation.

No coherent odd/even difference or secondary eclipse exactly half a cycle away emerges (Figure 3). An isolated sector-36 ingress-minus-egress contrast of -639 ± 233 ppm is retained rather than interpreted as a replicated asymmetry. A secondary at another phase, strong dilution, or an

unresolved companion could evade these checks. The descriptive depth means differ between the four older TESS-SPOC sectors (877 ± 47 ppm) and the later QLP sectors (599 ± 66 ppm). This is not established physical depth evolution. Same-epoch paired products give SPOC-minus-QLP differences of $+47 \pm 76$ ppm in sector 36 and -163 ± 96 ppm in sector 37, offering no consistent pipeline offset. Common-quality sampling itself changes fitted depths, and the later QLP extraction changed beginning in sector 94 Petitpas et al. (2026).

Time-averaging diagnostics motivated by Pont et al. (2006) increase the sector-100 error from 99 to 133 ppm. Its worst event-omission depth/error ratio is 2.96. These checks weaken the apparent precision without removing the signal; they do not calibrate its false-alarm probability. Four-sector orbital-harmonic tests likewise find no stable substantial modulation, but pipeline correction can suppress broad variability and this null result cannot exclude binaries.

Independent survey context is modestly favorable for the bright star. One securely associated RAVE observation on 2010 May 2 gives 5.94 ± 0.87 km s^{-1}, consistent with the Gaia aggregate velocity 7.02097 ± 0.32064 km s^{-1} at 1.17 combined formal error units. Predominantly normal RAVE morphology and dwarf-like atmospheric parameters do not constitute a companion exclusion. There is only one RAVE epoch, the surveys have different systematics, and a faint eclipsing neighbor would not be tested by this agreement. Gaia's available RV diagnostics provide no strong variability flag, but no orbital sensitivity or mass limit has been established.

4.2 TIC 402358350: localized recurrence with a failed timing gate

For TIC 402358350, the expanded clock frozen before sector 81 has $P = 11.769544332077816$ d, $t_0 = 2392.9928689397047$ BTJD, and $D = 0.144$ d (3.456 h). It incorporated sectors 40, 41, and 54; sector 54 had already been examined and is not an independent test of this expanded model. The original discovery clock was different and is retained in the evidence record.

The primary sector-81 prediction yields 634.6 ± 72.2 ppm with two supported events. However, its timing-profile spread of 0.09142 d exceeds the predeclared $D/2 = 0.072$ d gate. Consequently this is a positive dimming measurement with *inconclusive prediction timing*, despite a formal depth/error ratio of 8.79. The alternate expanded reduction and both original-clock reductions also fail timing precision. All four comparisons are shown in Figure 4; selecting the deepest comparison would not repair this failure.

A separate retrospective test directly fitted native SAP and PDCSAP photometry, normalized by each sector's median. Polynomial baselines of degree zero, one, and two used points outside the protected interval $|t - t_c| \leq D$, with outer radii $3D$ or $5D$. No running-median flattening or data-dependent clipping was added. Nine nominal centers fall within the sector spans; one sector-40 event lies in a gap and fails coverage for all twelve methods. The remaining eight events are positive under every method, yielding the 590–659 ppm combined depths in Table 7. Leaving out any one event retains 569–706 ppm across the matrix. Thus no individual dip or single baseline choice creates the measured recurrence.

The width tests show a localized depression rather than a uniformly broad low state (Figure 6). Adverse controls remain material: one matched SAP offset window is deeper than its associated event under five of six baselines, and a sector-54 event has no matched controls. Time-averaging inflation expands combined errors to 41–93 ppm, depending on baseline; linear-$3D$ results change only to 642 ± 43 and 610 ± 44 ppm. Scalar inflation is a sensitivity diagnostic rather than a full noise likelihood.

The nearby published pulsator TIC 402358249 has dominant frequency 0.9517724 d^{-1} (period 1.050671358 d). Adding sine and cosine nuisance terms at this fixed frequency leaves SAP depths near 578 ppm and PDCSAP depths near 610–621 ppm. This rules out only that simple sinusoidal

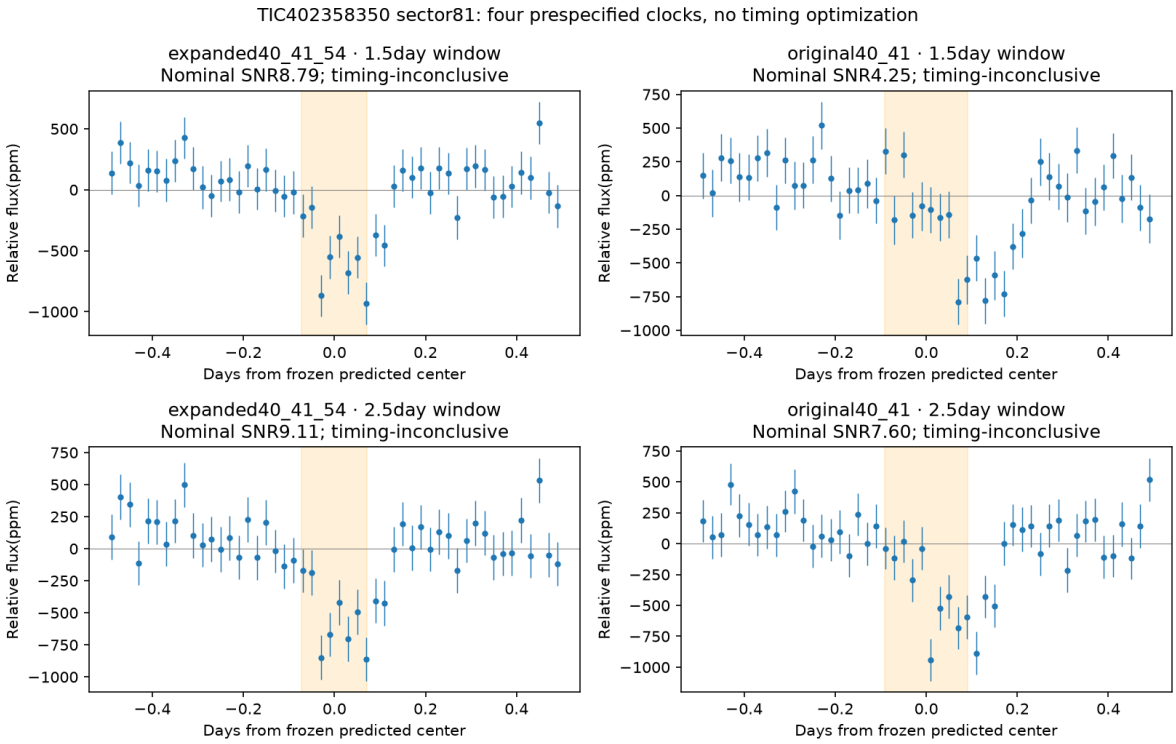


Figure 4: Sector-81 measurements for all four predictions fixed before access to TIC 402358350's sector-81 photometry. The expanded primary clock places a compact dip near the predicted center, but its timing spread exceeds the unchanged precision gate. The comparisons share the same sector-81 photons; neither alternate preprocessing nor a positive formal depth converts the result into a passed timing test.

Table 7: TIC 402358350: the complete twelve-method baseline matrix. Errors include baseline extrapolation uncertainty and residual-scatter scaling; correlations and source uncertainty remain incompletely modeled. SAP and PDCSAP are reductions of the same exposures.

Baseline	Outer radius	SAP depth (ppm)	PDCSAP depth (ppm)
Constant	$3D$	627 ± 44	609 ± 42
Constant	$5D$	602 ± 41	591 ± 40
Linear	$3D$	642 ± 41	610 ± 42
Linear	$5D$	614 ± 39	590 ± 39
Quadratic	$3D$	659 ± 53	647 ± 54
Quadratic	$5D$	633 ± 43	621 ± 43

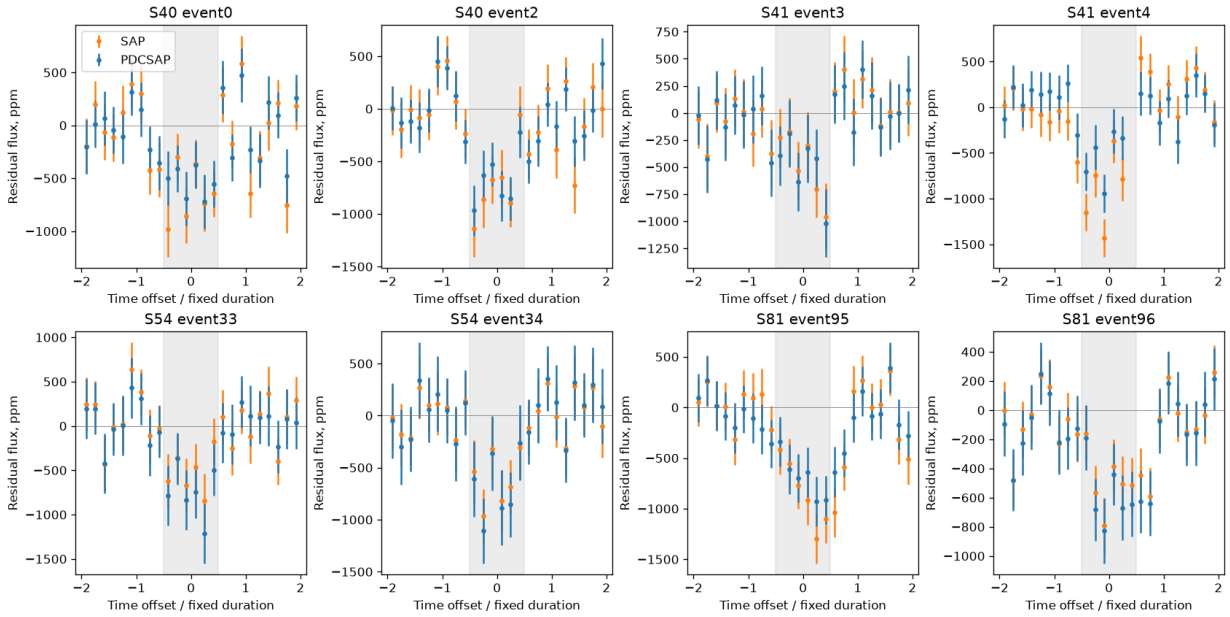


Figure 5: All eight usable TIC 402358350 events, illustrated with the linear-3D baseline specified in the method matrix. SAP and PDCSAP retain positive dips and share photons. The protected event intervals and local baselines test whether additional global smoothing created the feature; they do not establish the host. Errors follow the conditional measurement model and do not include all timing or instrumental uncertainty.

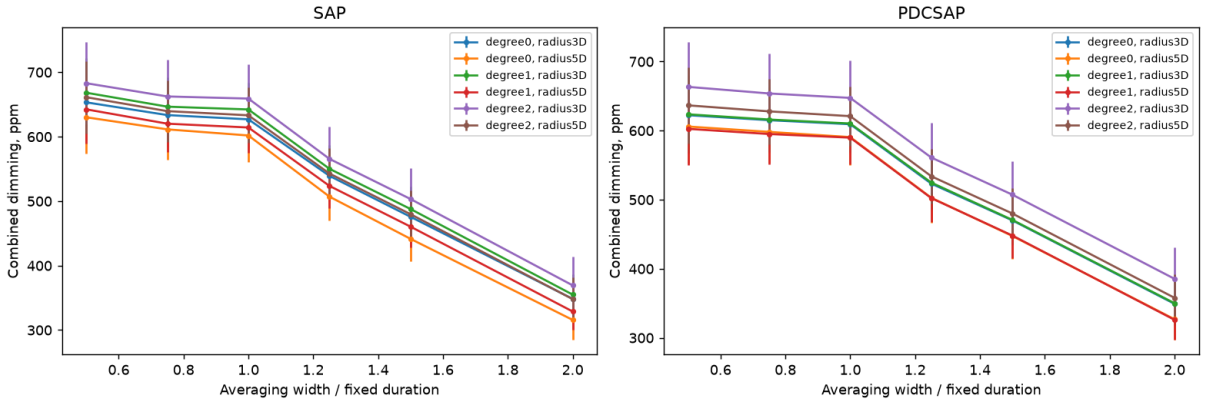


Figure 6: TIC 402358350's retained averaging-width and baseline matrix. Central depths are approximately stable between widths 0.5D and D, and decrease as wider windows include the shoulders. This supports localization in time; the displayed widths are fixed diagnostic choices, not a fitted transit-duration posterior. Rows and columns reuse the same events and are not independent trials.

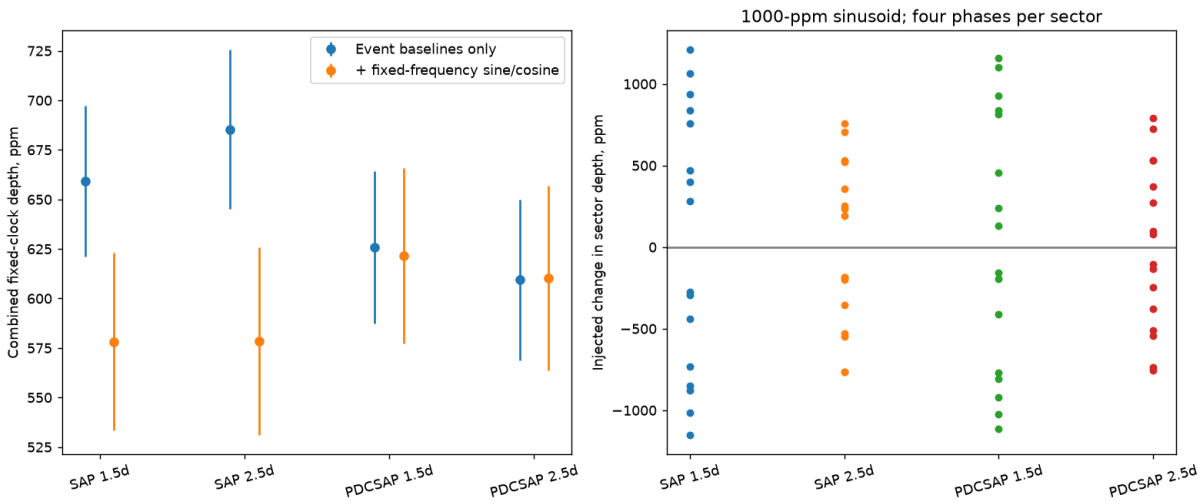


Figure 7: TIC 402358350's fixed-frequency pulsation nuisance fits and deterministic sinusoid-transfer tests. The nuisance sinusoid does not absorb the recurring dip, while the injection response demonstrates substantial sampling and preprocessing sensitivity. These are retrospective tests of already inspected products, not independent observations, an astrophysical contamination model, or a false-alarm calibration.

model as a complete explanation. Conversely, deterministic 1000-ppm sinusoidal injections into the archived products can shift a sector's box depth by up to 1209 ppm after sampling and flattening (Figure 7). Smooth variability can therefore couple strongly into these measurements; the experiment neither bounds the real contamination nor reruns the upstream telescope reduction.

5 Source attribution and competing stellar hosts

Repeatable dimming and identification of its host are separate requirements. For TIC 80387959, stacked image models prefer the target in sectors 36, 37, and 100, TIC 80387965 in sector 63, and TIC 80387961 in sector 90. Every sector changes its preferred host under at least one event omission (Figure 8). For TIC 402358350, the discovery sectors disagree, while later whole-stamp fits favor the target region with substantial structured residuals and sensitivity to the fitted image region. A known shallow planet, Kepler-93 b, is incorrectly assigned a neighbor preference by this same procedure in one tested sector. Consequently model rankings, even stable ones, are not calibrated host probabilities or sufficient planet vetoes.

The aperture-light budget supplies a complementary constraint Giacalone et al. (2022). If a neighbor contributes a fraction f of the aperture light, its intrinsic eclipse must have depth $e = d/f$ to produce an observed SAP decrement d. Under equal target/neighbor throughput, TIC 80387965, 13.76 arcsec away, needs approximately 1.41–1.72% eclipses; TIC 80387961, 7.54 arcsec away, needs 24.19–29.47%. For TIC 402358350, TIC 1918682563 at 10.12 arcsec needs 24.76–28.86%. These are conditional requirements, not observed eclipses. Some fainter competitors exceed 100% under nominal assumptions, but uncertain response, catalog identity, magnitude, and depth reopen them in the declared sensitivity tests (Figure 9). No listed close competitor is safely excluded, and unresolved companions are unconstrained by this resolved-source budget.

Resolved archival photometry has not closed this gap. For TIC 80387968, 35.34 arcsec from TIC 80387959, 43 quality-selected SkyMapper measurements include two differently colored samples during one predicted event, with no dimming. They disfavor a particular same-depth, flat-bottom eclipse at those instants, but fail the predeclared two-event exclusion gate. Color, eclipse shape,

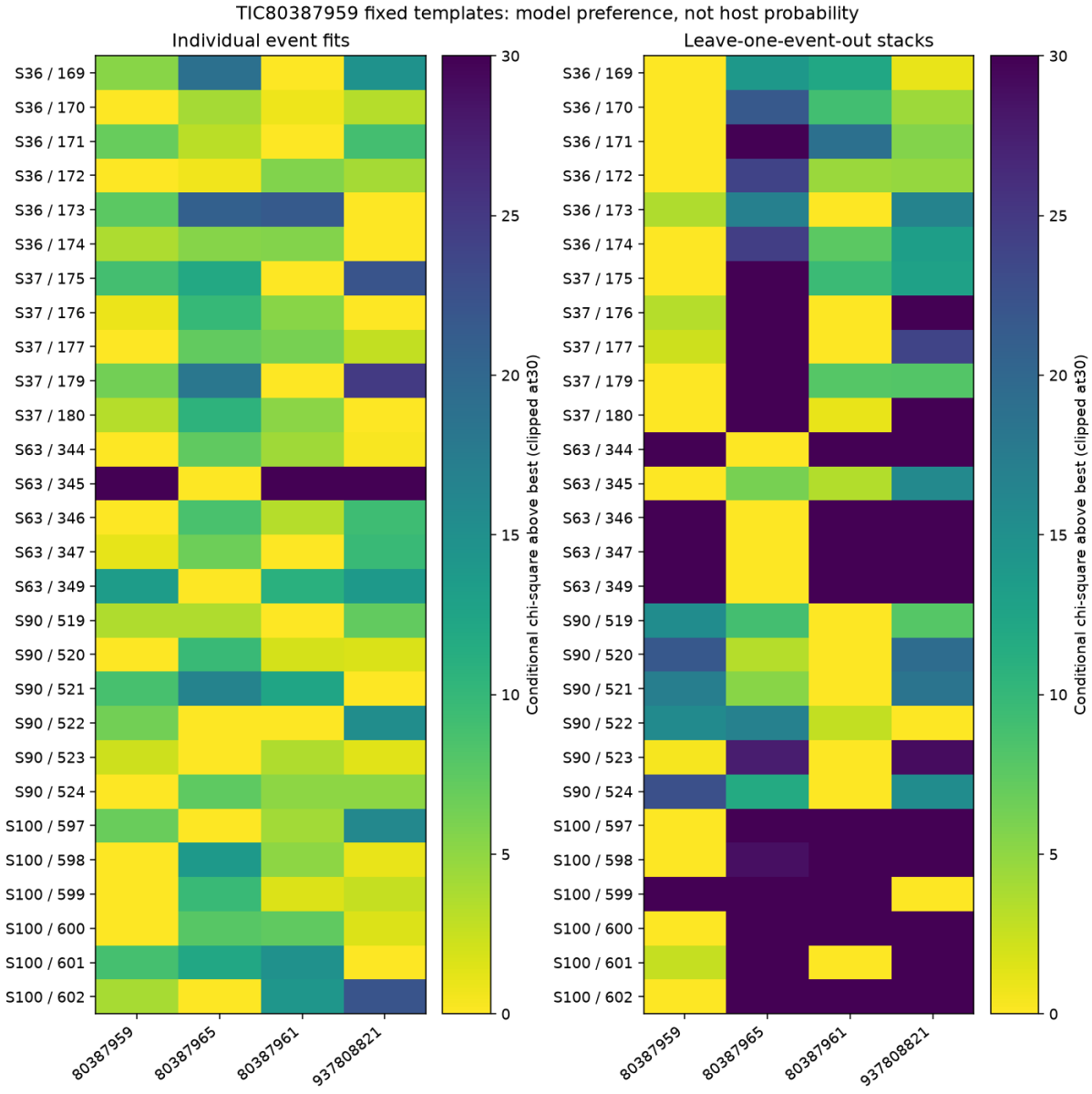


Figure 8: Event-level and omission sensitivity of TIC 80387959's source rankings. Different sectors and subsets prefer different stars. Overlapping omissions are correlated diagnostics. Inadequate image residuals and a failed known-host control prevent interpreting the chi-square rankings as probabilities or selecting target-favoring subsets as definitive source attribution.

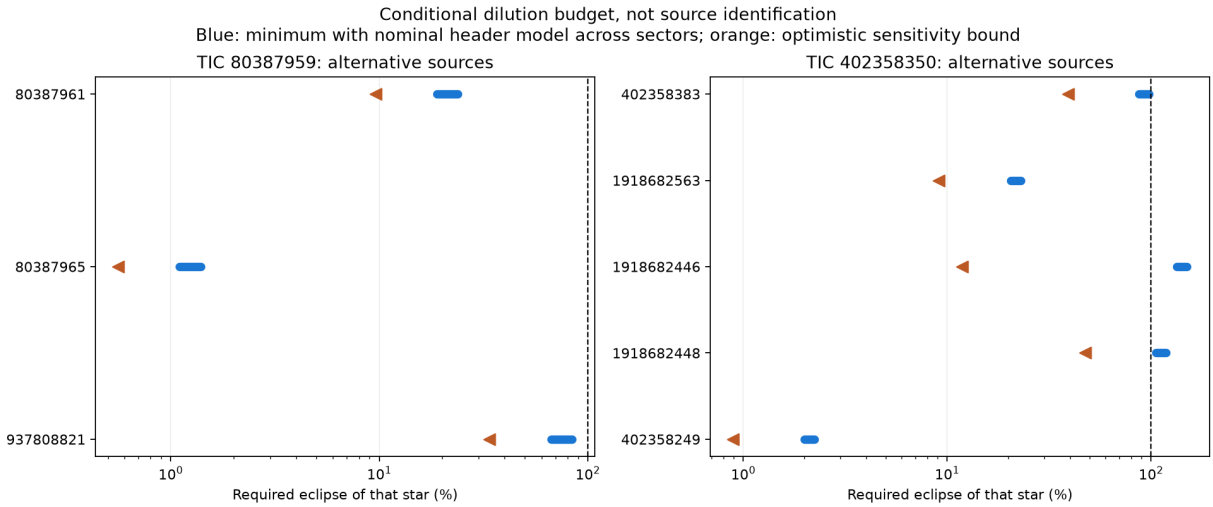


Figure 9: Conditional intrinsic eclipses required on nearby sources. Requirements use native SAP decrements and alternative aperture-response assumptions. Pipeline crowding and captured-flux estimates are models, not independent response measurements. The optimistic cases are declared stress tests rather than confidence limits. A nominal requirement above 100% does not exclude a source whose catalog or throughput model is uncertain.

and calibration remain limitations. Around TIC 402358350, four ZTF streams for TIC 1918682563 retain 288 accepted measurements with large scatter; all 48 fixed-clock comparisons are inconclusive. A separate six-stream test of TIC 74037550 retains 545 measurements and all 72 comparisons are inconclusive. Sparse sampling and propagated timing uncertainty prevent eclipse exclusions, even where nominal precision would otherwise appear sufficient.

6 Conditional astrophysical scenario calculations

We used TRICERATOPS 1.1.0 Giacalone et al. (2021) as an exploratory comparison of astrophysical scenarios, applying it to one already examined sector per target: sector 36 for TIC 80387959 and sector 54 for TIC 402358350. The native SAP inputs remove previously fitted event baselines, retaining 482 samples from six events and 230 samples from two events, respectively. The calculations use the diluted-flux convention and explicit exposure integration. Baseline covariance is not marginalized, stellar parameters are held fixed, and a nominal Gaussian aperture response controls nearby host eligibility. Verified exact catalog duplicates were removed while preserving the superseded inputs and calculations.

All 17 bounded runs and three provenance replays were completed. The nominal set contains three 100,000-draw seeds and one million-draw seed per target; the remaining runs test noise sensitivity and diagnostic controls. Table 8 gives the million-draw values. The original requirement of three independent million-draw calculations per target was not met, so convergence is unestablished. Three smaller seeds span 92.2334–92.7963% and 97.3632–97.8000% in target-primary weight; these ranges measure numerical scatter rather than physical uncertainty (Figure 10).

Here $1 - \mathrm{FPP}$ sums planet scenarios around the target primary, including the package's dilution alternatives. FPP includes some planets on other stars; NFPP includes all resolved-neighbor scenarios, including neighboring planets. Neither candidate meets both published numerical criteria, $\mathrm{FPP} < 0.015$ and $\mathrm{NFPP} < 0.001$; both also fail the nearby-source threshold for the method's less restrictive "likely planet" category. These thresholds do not independently transfer the calibration

Table 8: Single-seed, million-draw TRICERATOPS scenario weights. These are conditional on the supplied single-sector data and modeled scenarios; neither target passes the joint numerical validation thresholds.

TIC	Target-primary planet weight	FPP	NFPP
80387959	92.4284%	7.5716%	4.2862%
402358350	97.6834%	2.3166%	0.3959%

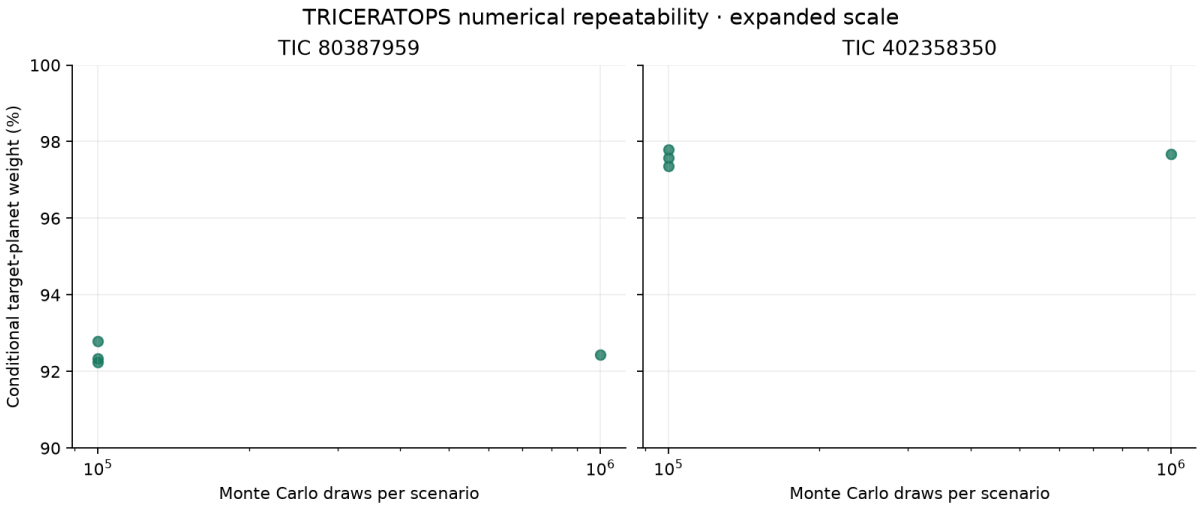


Figure 10: Adopted nominal TRICERATOPS runs at two draw counts. Repeated smaller calculations test Monte Carlo sensitivity, but one million-draw seed per target leaves the original convergence criterion unmet. The scenario weights exclude general artifacts and stellar variability, and do not quantify discovery reliability. The two targets also retain different observational timing and attribution limitations.

of the published workflow to the present exploratory inputs.

The omitted-model problem is visible in controls. Setting the flux identically to unity still yields target-primary planet weights of 44.15% and 23.38%; inverted signals yield 41.99% and 15.91%. These tests retain the supplied clock and observed-depth host eligibility, so they are computational diagnostics, not false-alarm experiments. Doubling the scalar noise changes nominal weights to 91.98% and 93.74%. The restricted Kepler-93 b control returns 82.88% and fails the validation thresholds despite that planet's independent establishment. The scenario family lacks general instrumental and stochastic stellar-variability alternatives and normalizes its weights even for unconvincing inputs. Accordingly these percentages do not measure the probability of a real planet or a new discovery. TIC 402358350's larger weight does not repair its timing failure or make its overall evidence stronger than that for TIC 80387959.

7 Four additional signals and their limitations

The remaining four signals illustrate distinct obstacles: a deep event with ambiguous companion size, two shallow signals with inadequate predictive timing, and recurring dimming for which a resolved neighboring binary is a plausible source. All measurements below retain the historical clocks and unfavorable alternatives. Formal errors describe the stated measurement model; they omit some correlated noise, timing and source uncertainties. Repeated reductions of one sector do not constitute independent observations.

7.1 TIC 241928959: target-region dimming with physical degeneracies

The approximately 3.533-day pattern is measured in sector 92 alone. Its refined training clock is $P = 3.532592898$ d with a 0.032-day box duration. Five usable, discovery-selected events give an event-baseline depth of $11\,992 \pm 786$ ppm; removing each event in turn leaves 11 729–12 442 ppm. One disturbed event remains excluded because its in-window samples carry nonzero quality flags and lack a usable baseline. The 2.5-day detrending removes another event's in-transit samples, leaving four observed transits. The physical fitter's six local windows include baseline-only windows and must not be counted as six transits. The deeper final event remains in the analysis.

Difference images favor the target region over TIC 241928961, a neighbor 10.56 arcsec away. A fuller scene model gives $\chi^2 = 109.16$ for the target and 437.05 for that neighbor, each with 117 degrees of freedom. All five individual events and their omission stacks prefer the target in the original brighter-scene model. These conditional template comparisons do not exclude unresolved companions or calibrate host probabilities. An illustrative equal-throughput blend with the known neighbor would require a roughly 24% eclipse; the actual aperture response is not known.

The short shape does not provide a secure companion radius (Figure 11). For the 1.5-day reduction, a circular, undiluted target model with freely fitted $a/R_\star$ gives radius ratio $k = 0.1257$, impact parameter $b = 0.840$ and an inferred stellar density 15.66 $\mathrm{g\,cm^{-3}}$. The nominal catalog mass and radius imply only about 3.21 $\mathrm{g\,cm^{-3}}$. A broad prior on $a/R_\star$ instead permits $k = 0.3072$ and $b = 1.135$, a grazing configuration, at a data-fit cost of only $\Delta\chi^2 \simeq 1.7$. The second detrending retains this degeneracy. Eccentricity, dilution and stellar-model errors remain unmodeled alternatives. The restricted neighbor fit reaches its upper radius-ratio bound; it is not a general eclipsing-binary exclusion.

Gaia RUWE 3.356 and a 14% multiple-peak image fraction reinforce caution about a single-star interpretation, without demonstrating a companion. Alternating depths differ by less than one formal standard deviation, but only one usable odd-family event is available. Fixed half-phase tests find no convincing secondary eclipse and do not exclude an eccentric or faint binary. Existing ZTF measurements are inconclusive, and the checked TESS coverage supplies no verified second observing epoch. Source identity, companion size and recurrence therefore remain unresolved.

7.2 TIC 469309982: mixed recurrence and a seasonal secondary concern

The four-sector weak-seed refinement uses sectors 9, 10, 36 and 37 and selects $P = 2.890464274$ d, a 0.144-day duration and a fitted depth of 138 ppm. Sector 64 at the unchanged nominal clock gives 75.8 ± 26.2 ppm. Its propagated timing spread, 0.12347 d, exceeds the allowed half-duration of 0.072 d. A previously saved alternative gives 133.4 ± 26.2 ppm, but this maximum across nine correlated clocks cannot replace the nominal prediction. Sector 63, evaluated using the clock selected after examining sector 64, is also weak: 82 ± 28 ppm, falling to 21 ppm when one event is omitted.

Only one of six retrospective season-out comparisons passes both the original recovery and timing conditions: the 1.5-day detrending predicts the omitted sectors 36/37 at 129.7 ± 22.6 ppm with a

0.0525-day timing spread (Figure 12). The corresponding 2.5-day result is positive, 136.7 ± 23.5 ppm, but separated accepted aliases expand its spread to 1.5706 d. All observations were previously inspected, the folds overlap, and the discovery seed uses sectors 9/10 even when that season is omitted.

A possible half-phase secondary is concentrated in sectors 36/37: 87.8 ± 22.7 ppm at the 1.5-day window, compared with 24.0 ± 22.6 and -0.5 ± 19.7 ppm in the earlier and later seasons. This remains a concern rather than a repeatable binary eclipse. Image models often favor neighbor TIC 469309967, but substantial residuals and failures on known-planet controls prevent a reliable source assignment.

Sector 90 gives -54.3 ± 31.6 ppm at the primary refined clock and 14.1 ± 33.8 ppm at the alternate refined clock. Positive measurements at the original unprofiled seed are retained without substitution (Figure 13). Timing is inadequate and expected formal sensitivity is below 5σ, so this does not establish either recovery or exclusion. The exact-source open-cluster reference supplies no useful age prior.[1]

7.3 TIC 96166003: positive depths depend on the clock

The original sectors 43/44 contain a compact approximately 0.955-day signal, with fixed-clock local depths near 280 ppm under both detrendings. Sector 19 provides only weak, timing-inconclusive measurements. Once it becomes training, the expanded primary clock is $P = 0.955530474$ d with duration 0.05952 d; the alternative selects $P = 0.955279111$ d and duration 0.048 d. All accepted profile islands remain preserved.

In sector 59 the primary predicts 264.9 ± 51.9 ppm, whereas the expanded alternative gives -72.4 ± 62.4 ppm and the two original-discovery comparisons also give negative depths. Every prediction fails its timing gate; the primary spread is 1.34665 d. Crossing both fixed clocks with both detrendings and SAP/ PDCSAP products retains the primary dip and the alternative's near-zero or negative result. Thus this disagreement follows the timing hypotheses more than these processing choices, but does not identify the correct hypothesis.

New sector 86 gives 248.2 ± 100.7 ppm at the expanded primary and 7.2 ± 134.0 ppm at its alternative. All six retained comparisons are low-sensitivity and timing-inconclusive (Figure 14). Discovery images favor the target nominally, but including faint neighbors changes some sector-43 rankings. A physically viable diluted-binary alternative remains. Neither the photometry nor template preferences securely establish a host. A bibliographic planet-table association is also uninformative.[2]

For each of these two shallow signals, bounded block controls give zero exceedances in 40 draws separately for each of three block models. Each conditional 95% Monte Carlo interval extends to 8.81%; counts are not pooled. These tests condition on selected targets and timing families and cannot supply survey false-alarm probabilities or planet odds.

7.4 TIC 5546257: resolved-neighbor variability strengthens contamination

The expanded pre-sector-62 clock, $P = 1.767290230$ d and duration 0.133056 d, gives 386.2 ± 30.1 ppm in sector 62. The timing spread, 0.07203 d, narrowly exceeds the allowed 0.06653 d. Newly opened QLP sectors 89 and 99 retain nominal clock dips of 271 ± 28 and 239 ± 33 ppm, but their timing

[1] TIC 469309982 is HD 300413. Its row in the checked Platais 8 membership table reports membership probability 0.00; bibliographic inclusion is not cluster membership.

[2] The exact TIC 96166003 row in the checked TESS yield paper describes a simulated 15.592607-day planet, not an observed planet. It neither validates nor invalidates the approximately 0.955-day signal investigated here.

spreads also fail the original gate. The coarse discovery seed gives negative depths. Different extraction methods and apertures prevent interpreting these depth changes as intrinsic evolution or improved source attribution.

Sector-62 source-plus-motion models favor neighboring TIC 5546242: $\chi^2 = 259.24/126$, compared with 776.24/126 for the target. All 13 event omissions and 10 of 13 individual images favor the neighbor, whose exact Gaia source is classified as an eclipsing binary. However, its catalog period, 7.248263937 TCB days, and the tested simple harmonics do not explain the selected TESS train. The closest period division, $P/4 = 1.812065956$ d, is 2.53% longer. This mismatch prevents claiming a complete catalog-clock explanation, while leaving contamination plausible.

Resolved Gaia epoch photometry of that neighbor supplies stronger adverse evidence (Figure 15). All five nominal in-window measurements at the unchanged refined TESS clock dim, with relative depth $16.343 \pm 0.192\%$; a sixth dim point lies just outside. The backward timing spread remains too large, and the 13 saved timing alternatives include only two to six of the conspicuous dim points. The published Gaia model explains most samples but leaves one notable discrepancy. Sparse sampling permits a possible catalog alias; no period was refitted to select that explanation. The neighbor association is strengthened, although its exact orbit, aperture dilution and quantitative explanation of the TESS signal remain unresolved.

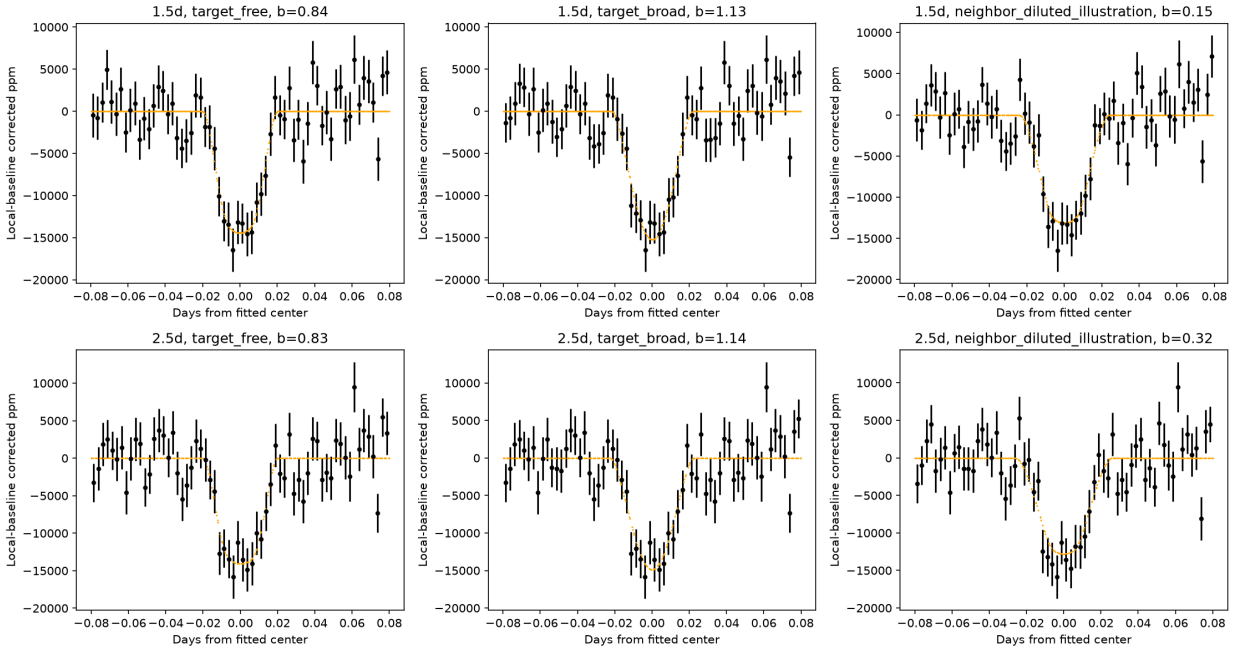


Figure 11: Conditional physical fits to the same sector-92 observations of TIC 241928959. Rows use the 1.5- and 2.5-day detrending windows; columns show a target model with free orbital scale, a target model with a broad scale prior, and an illustrative diluted-neighbor model. Similar folded shapes accommodate very different companion sizes and grazing geometries. The neighbor model reaches its radius-ratio bound and does not span general stellar-eclipse models. These fits supply no independently validated planet radius.

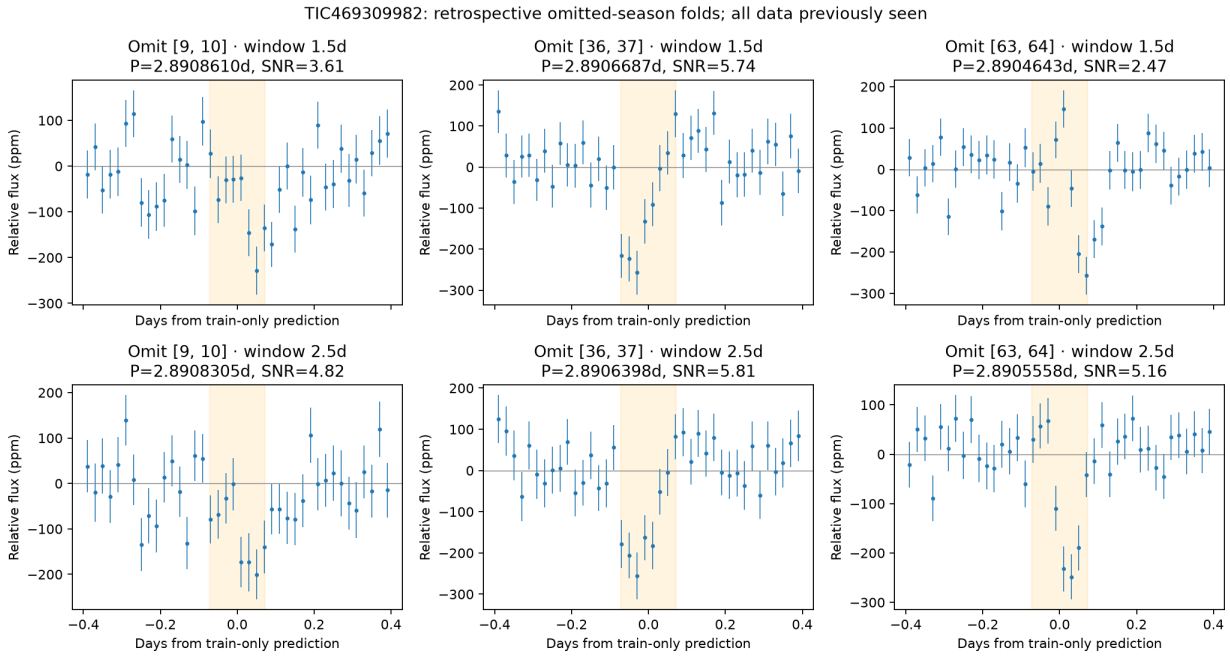


Figure 12: Retrospective omitted-season measurements for TIC 469309982. Columns omit sectors 9/10, 36/37 and 63/64; rows use two detrending windows. The shaded regions show the predicted boxes. Only the upper middle comparison passes both timing and recovery gates. The other five remain timing-inconclusive, including nominally strong lower-row features. All data were already inspected; the seed and overlapping training sets prevent interpreting these as independent blind discoveries. Displayed formal signal-to-noise values omit correlated noise and clock uncertainty.

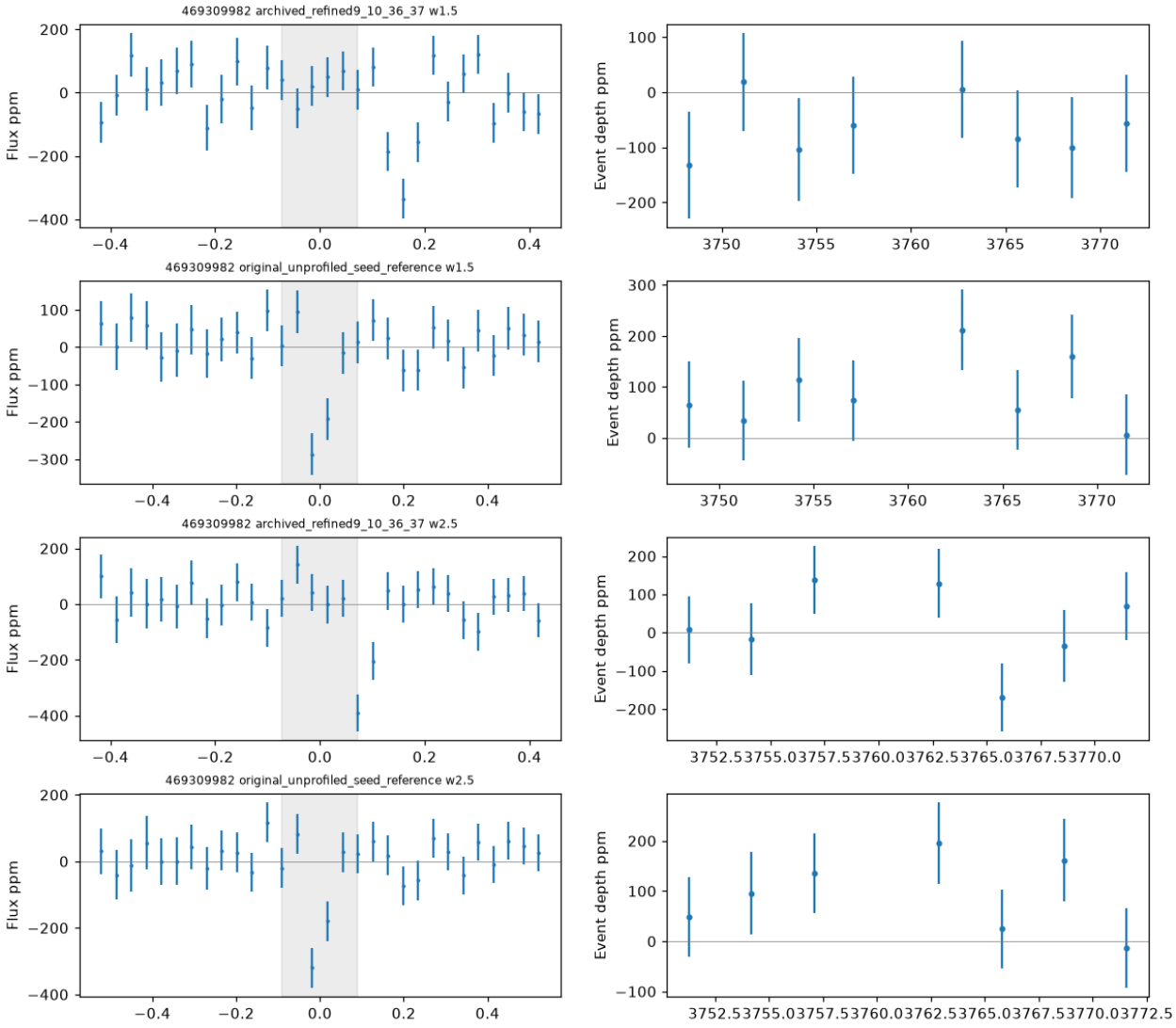


Figure 13: The four saved clock comparisons for TIC 469309982 sector 90. Left panels show fixed-clock folds (days from the predicted midpoint); right panels show individual-event depths at centers in BTJD. Positive depth denotes dimming. The primary result is negative. Profiled clocks fail timing precision, unprofiled seed references are ineligible for recovery, and expected formal sensitivity is below 5σ in every row. This reproduces the upper four rows of the original ten-row chart; the lower six rows are shown separately in Figure 14 for readability.

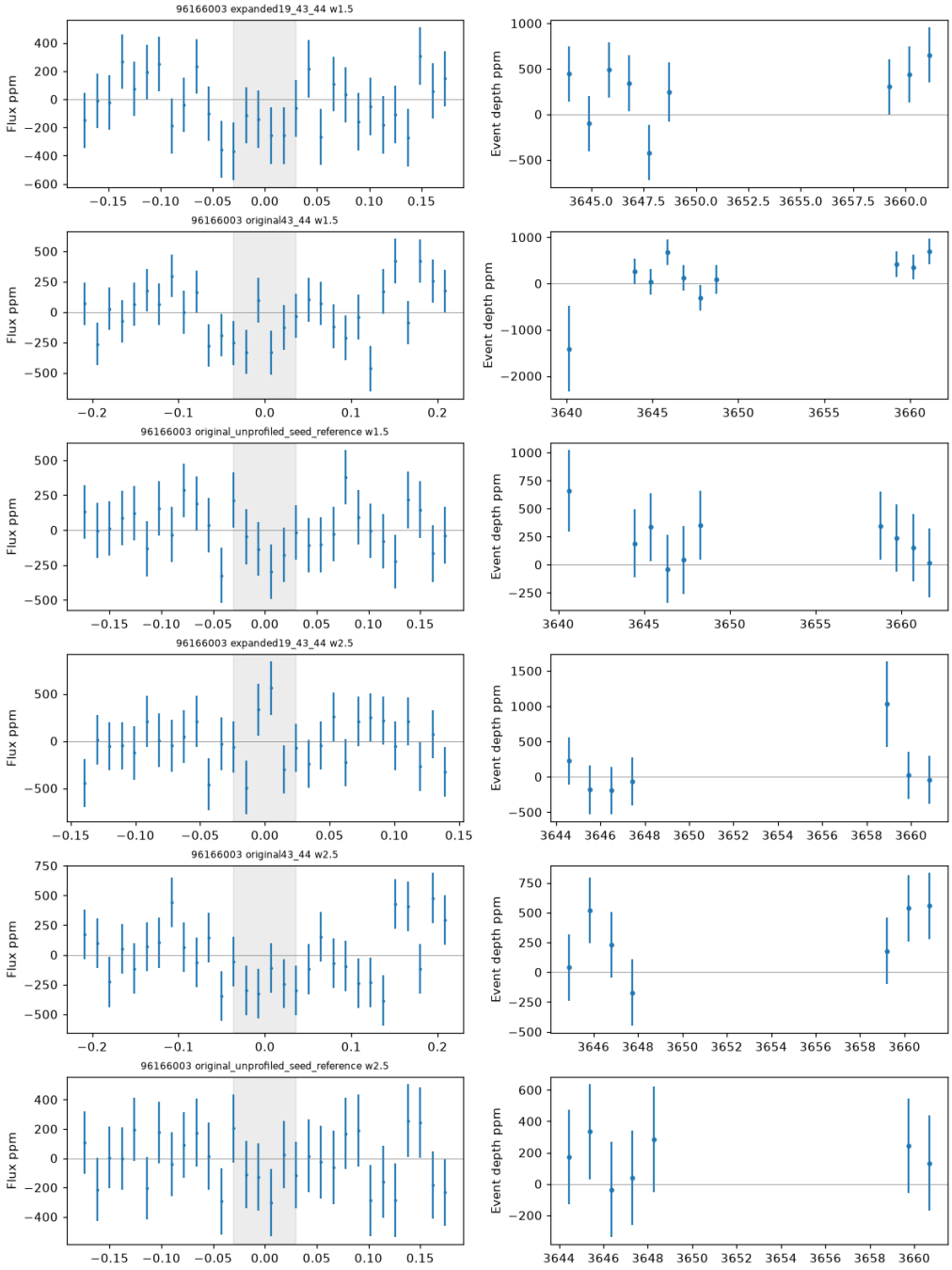


Figure 14: The six saved clock comparisons for TIC 96166003 sector 86, reproducing the lower six rows of the same original chart as Figure 13. The primary is weak, and the expanded alternative is near zero. Left abscissae are days from the predicted midpoint; right abscissae are event centers in BTJD. Profiled clocks fail timing precision, unprofiled seed references are ineligible for recovery, and expected formal sensitivity is below 5σ in every row. Repeated reductions share photons; none passes the original recovery requirements. Only page layout has changed.

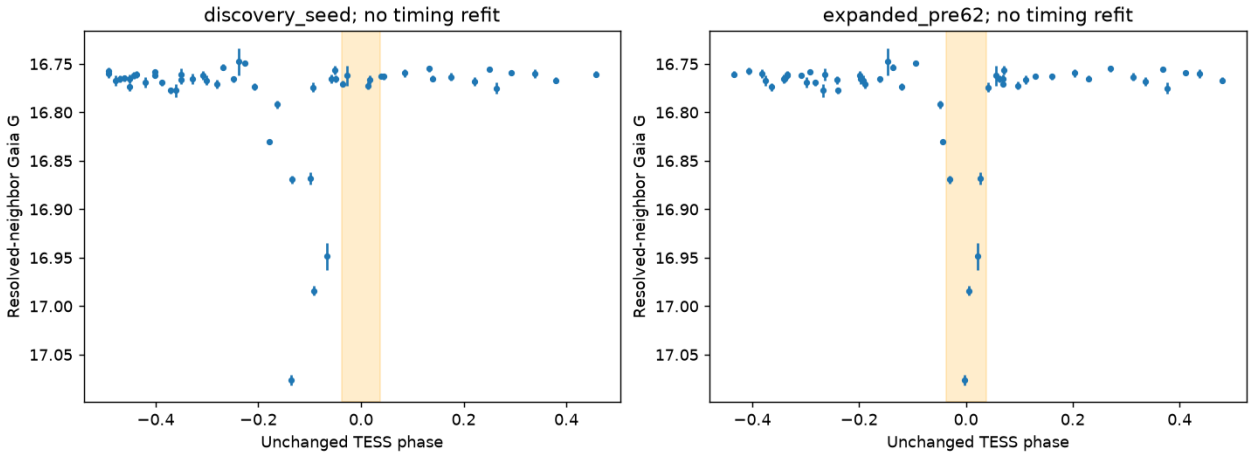


Figure 15: Resolved Gaia G measurements of TIC 5546242, the binary neighbor favored by TIC 5546257 difference images. Left: unchanged coarse discovery clock. Right: unchanged expanded pre-sector-62 clock. The latter aligns five dim points within the frozen window and a sixth nearby, supporting contamination. The backward timing gate still fails, and alternate saved profiles do not all align the same points. Neither this nominal alignment nor the catalog model settles the period alias or the TESS dilution. These are neighbor measurements, not independently confirmed target transits.

8 Interpretation and follow-up

The evidence supports different levels of recurrence, without completing the chain from a selected dip to a planetary discovery. TIC 80387959 has the clearest prospective recurrence: its fixed earlier clock recovered dimming in two later sectors, and retrospective native-cadence fits preserve the dip across six sectors. TIC 402358350 has the strongest local-baseline stress test, but its later-sector timing gate fails. The other fields preserve useful patterns or contamination evidence with weaker independent recurrence. These distinctions should guide follow-up rather than be collapsed into a numerical ranking of planet probabilities.

Several checks constrain only particular alternatives. A missing exact-half-phase secondary does not exclude an eccentric or diluted stellar binary. A sinusoidal nuisance model does not span general stellar variability. A normal single spectrum or two consistent survey velocities does not exclude an eclipsing companion. Sparse photometry outside the relevant timing envelope cannot veto an eclipse. Catalog non-matches and failed archive queries must remain separate: neither proves that a star is constant, single, or previously unstudied. Qualitative follow-up priority and agreement among computational checks are not independent validation.

Resolved, time-resolved photometry is the most direct next experiment for the two priority fields. It should measure the target and all viable neighbors through the entire predicted event with sufficient out-of-event baseline. A target-centered dimming would still require constraints on unresolved companions. Appropriate spectroscopy and high-resolution imaging could then reduce blend and stellar-companion ambiguity, as in established validation workflows (Giacalone et al., 2022; Cacciapuoti et al., 2022; Eisner et al., 2021). Candidate-specific error and throughput calibration are necessary before interpreting an image-model preference as a source probability.

The retained planning packet contains 47 nominal TIC 80387959 windows and 17 nominal TIC 402358350 windows through April 2027. These are predictions, not observed events or telescope allocations. Selected-profile timing envelopes plus one hour of baseline on each side require approximately 5.6 hours for TIC 80387959 and 17.5–18.5 hours for TIC 402358350. They are heuristic timing ranges, not calibrated confidence bounds, and older alternative clocks can imply larger uncertainty. No full conservative LCO one-meter track passes the retained mount-margin assessment. One T17 date passes the tested sky geometry, but the actual instrument horizon, mount constraints, access, and precision remain unverified. No observing time is secured and no inquiry has been sent. A short non-detection under these conditions would be inconclusive.

9 Conclusions

We retain six unresolved TESS signal assessments and an explicit record of the tests that favor or weaken their interpretations. The fixed-clock later-sector recovery of TIC 80387959 and the eight-event baseline robustness of TIC 402358350 justify additional investigation. Their host stars and physical causes remain uncertain. TIC 241928959 lacks a second epoch and permits substantially different companion interpretations; TIC 469309982 and TIC 96166003 have weak or clock-dependent recurrence; TIC 5546257 has strengthened evidence for a neighboring eclipsing source without a complete alias and dilution explanation. None is validated or confirmed as a planet, and novelty is unestablished. The value of the present release is a reproducible starting point for decisive follow-up, including the negative evidence needed to design it.

Data, code, and provenance

The code, saved measurements, figures, run plans, input manifests, and verification reports are available at https://github.com/relh/exoplanet. This manuscript uses the scientific state assessed on 13 October 2026; its preparation did not open new observing epochs or run a new candidate search. The public evidence baseline is commit `0537d585d`. Appendix A links the principal reports. The accompanying `evidence-manifest.json` records full file paths and SHA-256 hashes for every reproduced figure and the selected numerical, plan, source, and assessment records. The standalone source package includes the figures needed to compile this paper; the full repository retains the larger evidence archive and third-party data credits.

The publication cleanup removed operational host/session metadata and authentication parameters from expired signed download URLs. Original frozen-file hashes are retained, with original-to-sanitized mappings in `docs/privacy-edits.json` and `docs/release-history-redactions.json`. Edited plans are not new pre-access freezes. Exact original-byte verification can require the translation or the retained private originals; numerical experiment fields and scientific conclusions were preserved. Raw downloads remain outside Git and are identified by the run manifests. Reproducing arithmetic from a saved dataset does not independently validate its scientific model.

Acknowledgments and software

This work uses public TESS observations, TESS-SPOC and QLP products from MAST, ESA Gaia data processed by the Gaia Data Processing and Analysis Consortium, and the external surveys credited in the individual reports, including RAVE, SkyMapper, and ZTF. TESS mission funding is provided by NASA's Science Mission Directorate. The TESS-SPOC dataset is identified by doi:10.17909/t9-wpz1-8s54; product-level identities and retrieval hashes remain in the research archive. Third-party material retains its original terms and attribution. Software includes Python, NumPy, SciPy, Matplotlib, Astropy (Astropy Collaboration et al., 2022), batman (Kreidberg, 2015), and TRICERATOPS (Giacalone et al., 2021). AI-assisted tools were used in preparing this manuscript; their output and agreement are not independent scientific validation. The author is responsible for the reported interpretations.

References

Astropy Collaboration, Adrian M. Price-Whelan, et al. The Astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package. *Astrophysical Journal*, 935:167, 2022. doi: 10.3847/1538-4357/ac7c74.

Luca Cacciapuoti et al. The TESS triple-9 catalog: 999 uniformly vetted candidate exoplanets. *Monthly Notices of the Royal Astronomical Society*, 513:102–116, 2022. doi: 10.1093/mnras/stac652.

Douglas A. Caldwell et al. TESS science processing operations center FFI target list products. arXiv:2011.05495, 2020. URL https://arxiv.org/abs/2011.05495.

Nora L. Eisner et al. Planet hunters TESS II: findings from the first two years of TESS. *Monthly Notices of the Royal Astronomical Society*, 501:4669–4690, 2021. doi: 10.1093/mnras/staa3739.

Gaia Collaboration, A. Vallenari, et al. Gaia data release 3: Summary of the content and survey properties. *Astronomy and Astrophysics*, 674:A1, 2023. doi: 10.1051/0004-6361/202243940.

Steven Giacalone et al. Vetting of 384 TESS objects of interest with TRICERATOPS and statistical validation of 12 planet candidates. *Astronomical Journal*, 161:24, 2021. doi: 10.3847/1538-3881/abc6af.

Steven Giacalone et al. Validation of 13 hot and potentially terrestrial TESS planets. *Astronomical Journal*, 163:99, 2022. doi: 10.3847/1538-3881/ac4334.

Chelsea X. Huang et al. Photometry of 10 million stars from the first two years of TESS full frame images. arXiv:2011.06459, 2020. URL https://arxiv.org/abs/2011.06459.

G. Kovács, S. Zucker, and T. Mazeh. A box-fitting algorithm in the search for periodic transits. *Astronomy and Astrophysics*, 391:369–377, 2002. doi: 10.1051/0004-6361:20020802.

Laura Kreidberg. batman: BAsic Transit Model cAlculatioN in Python. *Publications of the Astronomical Society of the Pacific*, 127:1161–1165, 2015. doi: 10.1086/683602.

Glen Petitpas et al. QLP data release notes 004: TESS-Gaia light curve photometry implementation. arXiv:2603.22236, 2026. URL https://arxiv.org/abs/2603.22236.

F. Pont, S. Zucker, and D. Queloz. The effect of red noise on planetary transit detection. *Monthly Notices of the Royal Astronomical Society*, 373:231–242, 2006. doi: 10.1111/j.1365-2966.2006.11012.x.

George R. Ricker et al. Transiting exoplanet survey satellite (TESS). *Journal of Astronomical Telescopes, Instruments, and Systems*, 1:014003, 2015. doi: 10.1117/1.JATIS.1.1.014003.

Keivan G. Stassun et al. The revised TESS input catalog and candidate target list. *Astronomical Journal*, 158:138, 2019. doi: 10.3847/1538-3881/ab3467.

A Evidence inventory

The following reports provide the complete measurements, alternatives, quality failures, and run-specific verification. Links are pinned to the public evidence baseline. File hashes and figure-to-source mappings are in the accompanying manifest; the reports link the underlying plans, numerical arrays, and source snapshots. Historical conclusions are superseded only where the current assessment says so.

ID	Evidence	Repository path
E01	Scope and evidence index	CHECKED-RESULTS.md
E02	Search completion	results/scoped-completion-20261012/ASSESSMENT.md
E03	Solar selection	results/solar-radius-pilot-selection-20261011/ASSESSMENT.md
E04	Solar ledger	results/solar-radius-lead-audit-20261011/ASSESSMENT.md
E05	Weak-seed review	results/weak-seed-audit-20261011/ASSESSMENT.md
E06	Faint QLP expansion	results/qlp-faint-expanded-20261010/ASSESSMENT.md
E07	Reserved-data exception	results/solar-radius-ranks3920-4432-20261011/ASSESSMENT.md
E08	Known-host controls	results/solar-radius-positive-controls-20261011/ASSESSMENT.md
E09	Shallow injections	results/solar-radius-shallow-injections-20261011/ASSESSMENT.md
E10	803 frozen prediction	results/followup-80387959-qlp90-100-20261009/heldout.json
E11	803 native profiles	results/strongest-80387959-shape-20261012/ASSESSMENT.md
E12	402 frozen sector 81	results/heldout-402358350-s81-20261011/ASSESSMENT.md
E13	402 local baselines	results/strongest-402358350-baselines-20261012/ASSESSMENT.md
E14	803 noise audit	results/paper-vetting-80387959-20261012/ASSESSMENT.md
E15	402 noise audit	results/paper-vetting-402358350-20261012/ASSESSMENT.md
E16	Neighbor light budgets	results/paper-host-vetting-20261012/ASSESSMENT.md
E17	Image control failure	results/prf-137151335-20261011/ASSESSMENT.md
E18	Source instability	results/prf-80387959-event-stability-20261011/ASSESSMENT.md
E19	Pointing controls	results/pointing-correction-controls-20261012/ASSESSMENT.md
E20	Aperture controls	results/strongest-host-identifiability-20261012/ASSESSMENT.md
E21	TRICERATOPS results	results/triceratops-20261013/ASSESSMENT.md
E22	TRICERATOPS review	results/triceratops-review-20261013/ASSESSMENT.md
E23	Resolved SkyMapper	results/resolved-followup-803-20261013/ASSESSMENT.md
E24	Resolved ZTF	results/resolved-followup-402-20261013/ASSESSMENT.md
E25	241 physical fits	results/physical-241928959-20261012/ASSESSMENT.md
E26	469 season tests	results/season-out-469309982-20261011/ASSESSMENT.md
E27	469/961 new epochs	results/new-epochs-469309982-96166003-20261012/ASSESSMENT.md
E28	961 clock comparison	results/same-clock-96166003-20261012/ASSESSMENT.md
E29	554 new epochs	results/new-epochs-5546257-20261012/ASSESSMENT.md
E30	Pulsator and Gaia tests	results/extension-402358350-5546257-20261012/ASSESSMENT.md
E31	554 catalog alias	results/contaminant-clock-5546257-20261012/ASSESSMENT.md
E32	Six local baselines	results/event-baseline-six-20261012/ASSESSMENT.md
E33	Companion metadata	results/companion-constraints-shortlist-20261012/ASSESSMENT.md
E34	803 spectroscopy	results/web-80387959-20261012/ASSESSMENT.md
E35	Follow-up planning	results/followup-planning-20261013/ASSESSMENT.md
E36	Observing access	results/observing-access-20261013/ASSESSMENT.md

B Additional diagnostic figures

These figures retain noise, source-identification, external-photometry, and processing limitations alongside the primary light-curve evidence. Panels from the same exposures are dependent comparisons. Their error bars do not include all source, timing, or instrumental uncertainties.

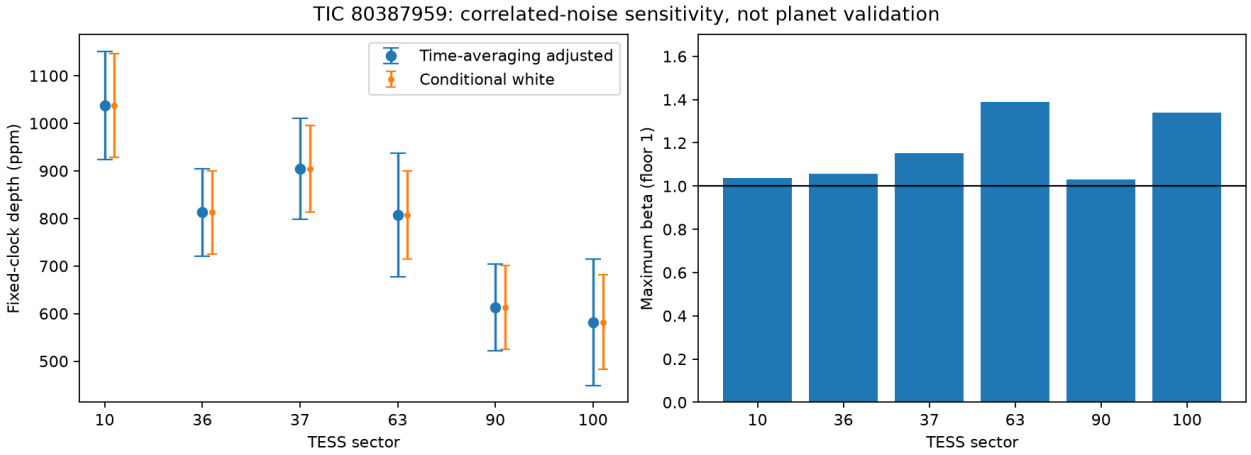


Figure 16: TIC 80387959 time-averaging sensitivity. The six sector depths are retained while approximate error inflation and individual-event omissions reduce their apparent precision. The white-noise reference uses the same fitting design. Scalar inflation is not a complete correlated-noise model or a calibrated detection significance.

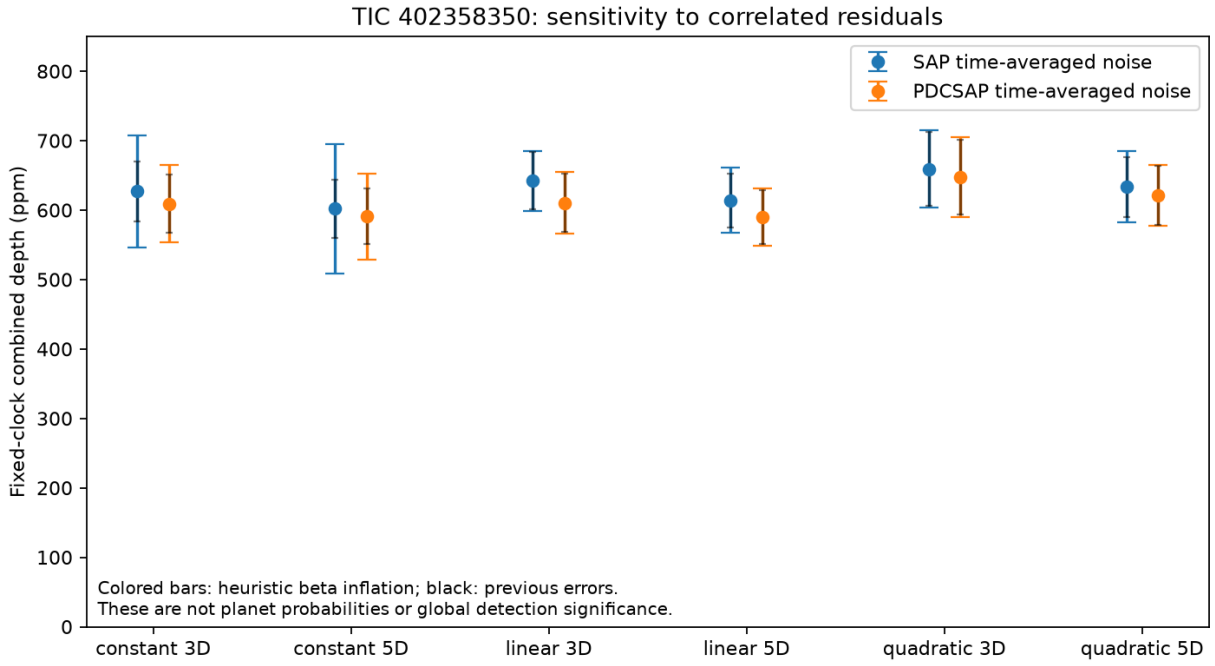


Figure 17: TIC 402358350 noise sensitivity across the twelve SAP/PDC baseline methods. Missing or insufficiently populated time bins remain unavailable; the combined error range is 41–93 ppm. Reweighting the events is a separately labeled sensitivity test. These methods reuse the same eight events.

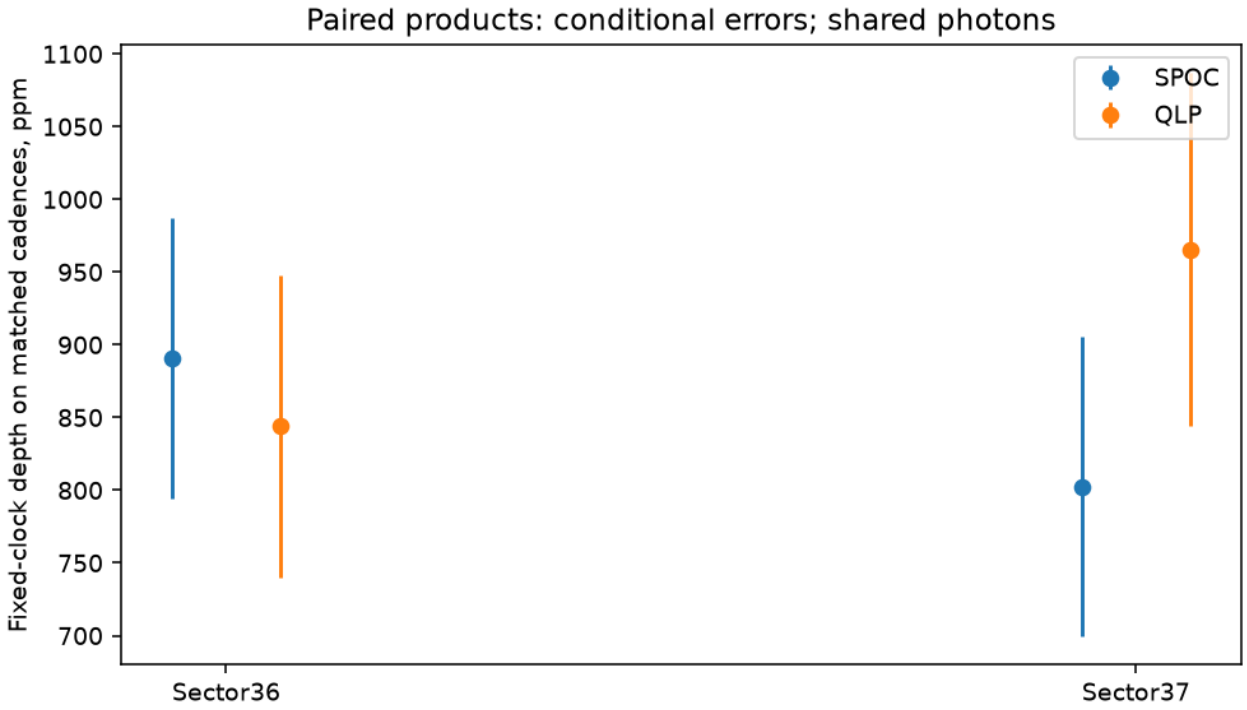


Figure 18: TIC 80387959 paired SPOC/QLP reductions of the same sector-36/37 exposures. Differences depend on epoch and common-quality selection. They do not establish a universal QLP depth suppression or physical depth evolution.

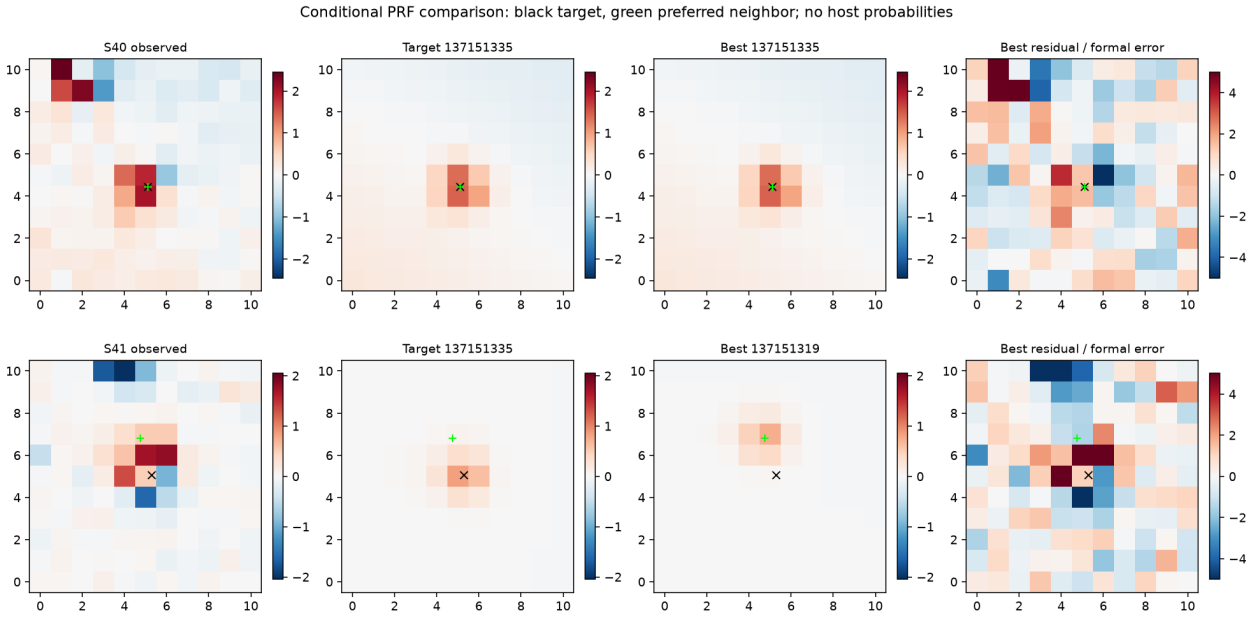


Figure 19: Known-host Kepler-93 b image control. The original template method prefers the correct target in sector 40 and a neighbor in sector 41 despite the established planetary signal. Structured residuals and wrong rankings undermine a probability interpretation of the same method on the candidates.

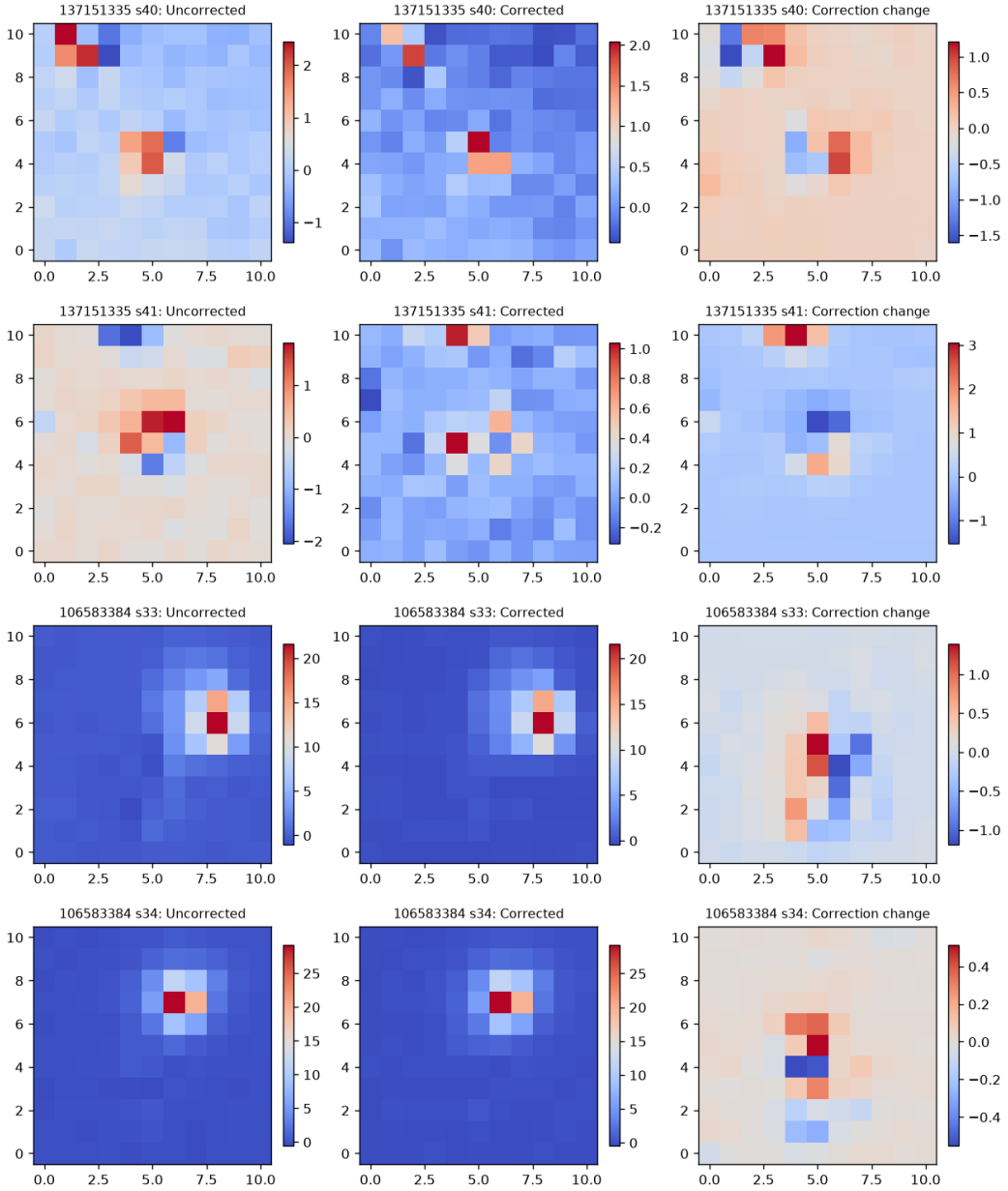


Figure 20: Pointing-correction controls: three of four satisfy the frozen criterion. The Kepler-93 wrong-host case improves, but discrimination decreases in one contaminant sector. The complete gate therefore fails and the method is not applied to infer candidate hosts.

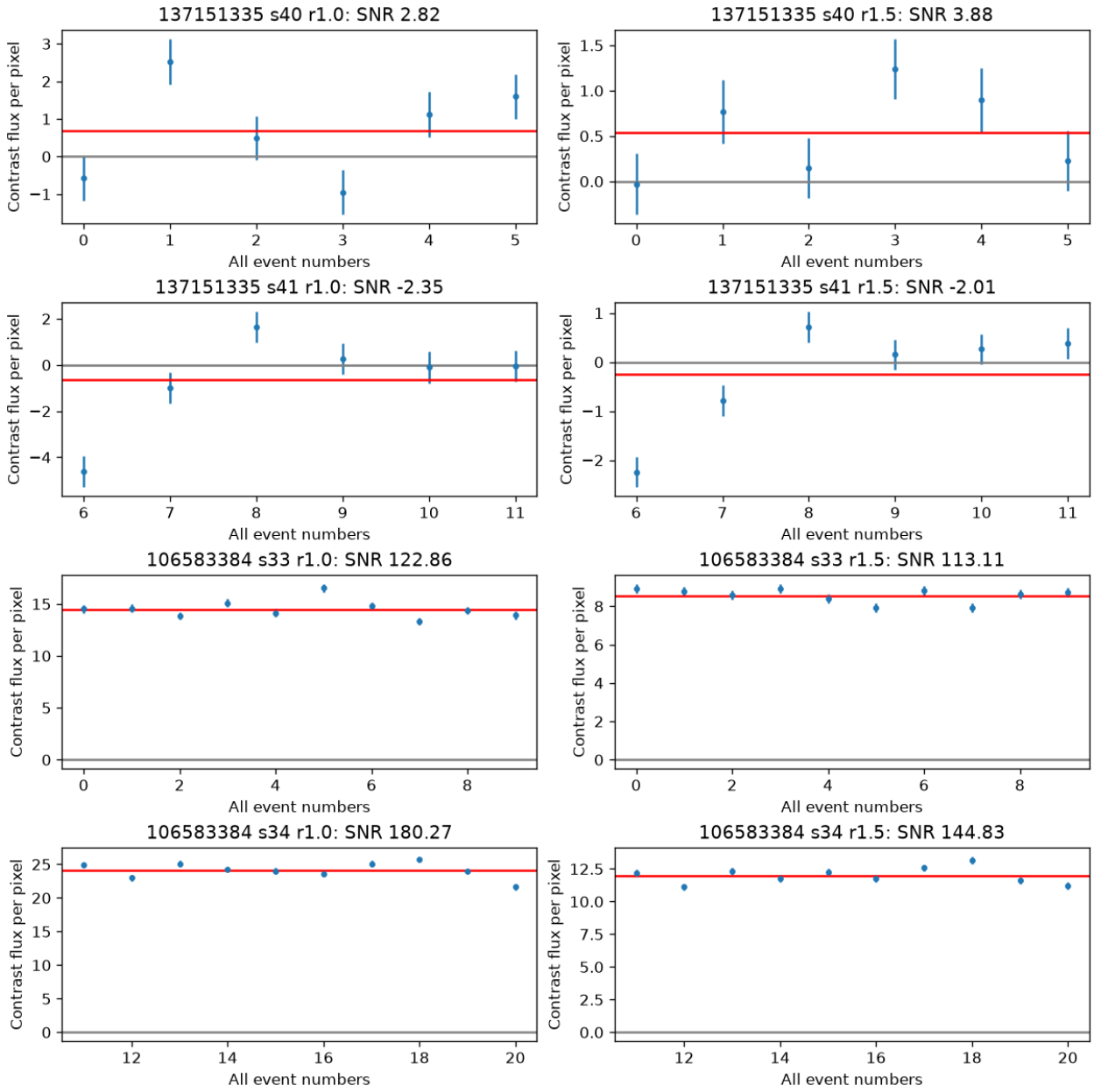


Figure 21: Paired-aperture controls: five of eight sector/radius cases pass. Kepler-93 failures include a weak target measurement and wrong-region contrasts. The ordinate measures differential variable flux per exclusive pixel, not stellar fractional eclipse depth or calibrated throughput. The candidate-application gate fails.

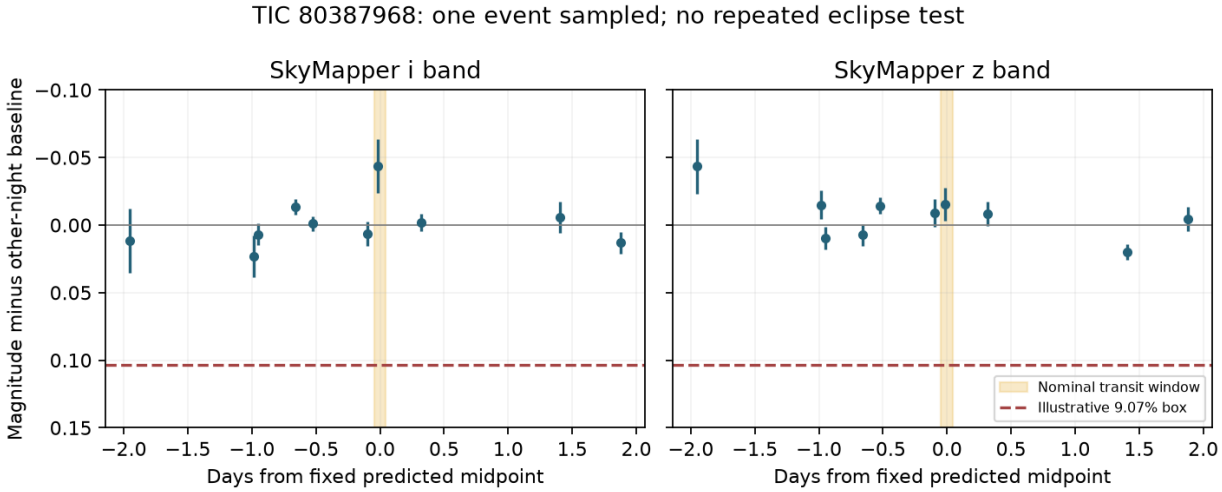


Figure 22: Resolved SkyMapper samples for neighbor TIC 80387968. Two measurements in different filters sample one predicted event without dimming, weakening a specified same-depth eclipse at those instants. The required two-event exclusion gate fails; sparse color-dependent samples do not clear the neighbor.

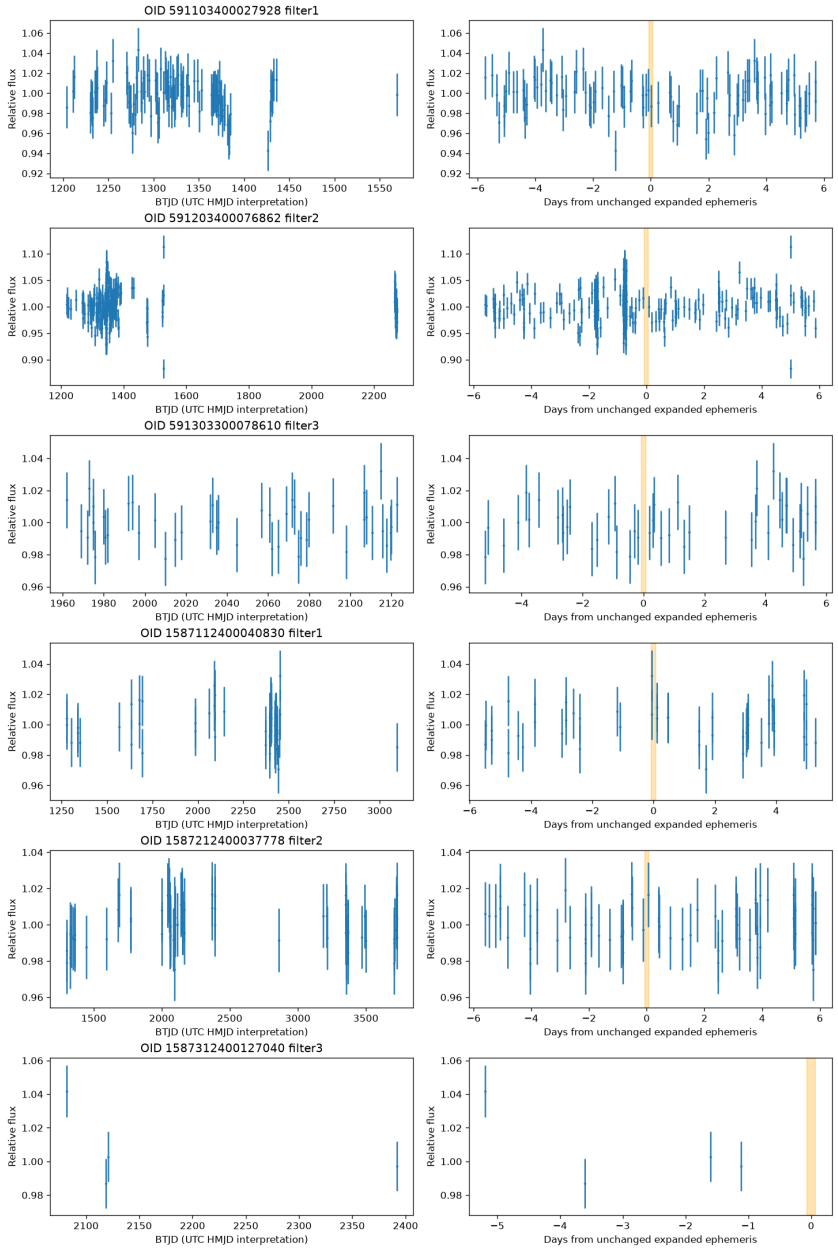


Figure 23: Resolved ZTF measurements for neighbor TIC 74037550 near TIC 402358350. Six streams contain 836 retrieved and 545 quality-selected points. All 72 clock/time-convention/stream-or-filter comparisons remain inconclusive; these plots provide neither a recovered eclipse nor a neighbor exclusion.

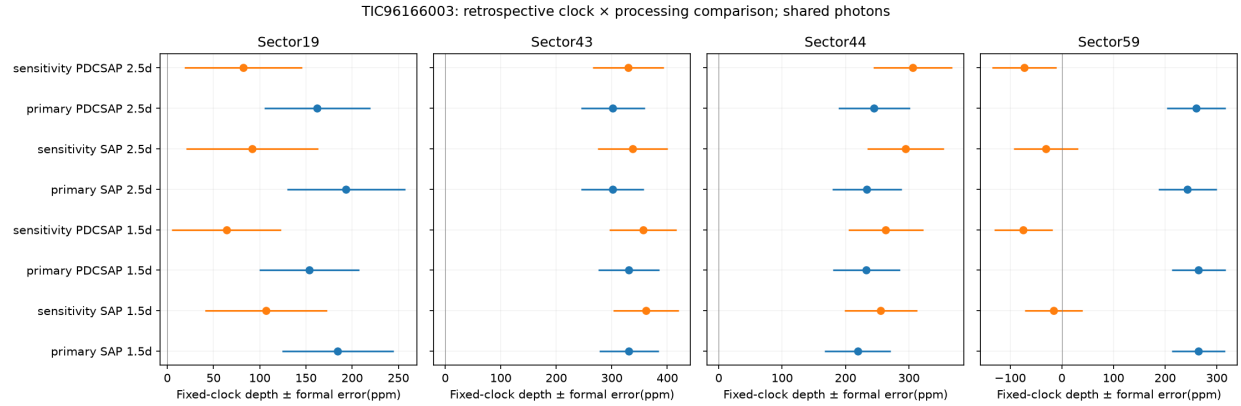


Figure 24: Crossed clock and processing comparison for TIC 96166003 sector 59. The expanded primary clock retains dimming across SAP/PDC and two detrending choices, whereas its alternative is negative or near zero. Shared-photon processing agreement does not identify the correct clock, and the original prediction timing remains inadequate.

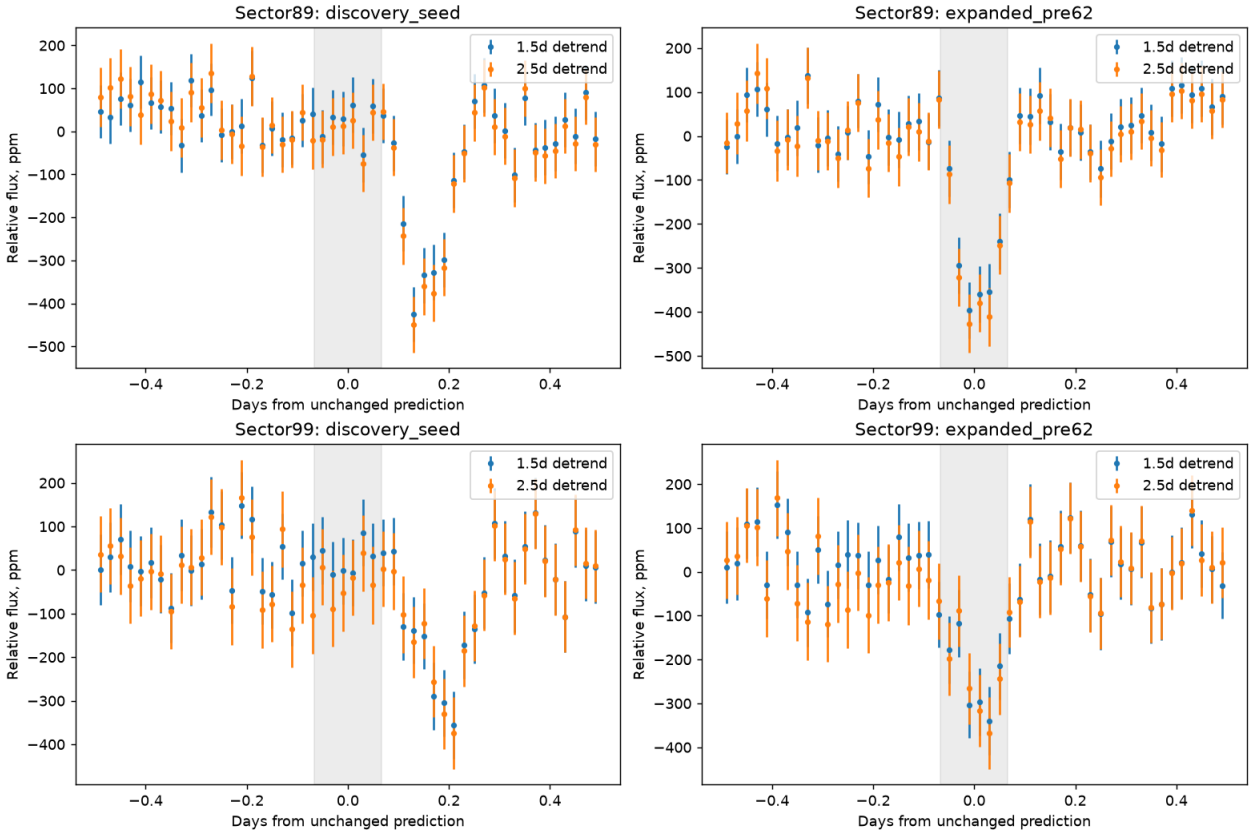


Figure 25: TIC 5546257 sectors 89 and 99 at the saved clock alternatives. Expanded-clock dips remain positive but fail the original timing gate; the coarse seed is unfavorable. Extraction and aperture differences limit physical depth comparisons. Neighbor Gaia variability strengthens contamination independently of these blended reductions.